SingleRAN

Power Supply Management Feature Parameter Description

Issue Draft B

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Contents

1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes

  Changes in functions and their corresponding parameters

- Editorial changes

  Improvements or revisions to the documentation

1.1 SRAN22.1 Draft B (2026-01-30)

This issue includes the following changes.

Technical Changes

None

Editorial Changes

Revised descriptions in the document.

1.2 SRAN22.1 Draft A (2025-12-31)

This issue introduces the following changes to SRAN21.1 01 (2025-03-10).

Technical Changes

Change Description	Parameter Change	Base Station Model
Enhanced basic MTS O&M on the MAE. For details, see: • **4.4.1.1 Data Preparation** • **4.4.1.2.1 Activation Command Examples** • **4.4.2 Activation Verification** • **4.5.1.1 Data Preparation** • **4.6 Operation and Maintenance (Multimode Base Station)** • **13.3.3 Hardware** • **13.3.4 Others**	Added the **DSP BATTERYITEM** command. Modified parameter: Modified the value range of the **PMU**.***DCVUTHD (LTE eNodeB, 5G gNodeB)*** parameter to 530–600.	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added the iLiBattScan feature. For details, see: • **2.3 Features in This Document** • **7 iLiBattScan**	Added parameters: • **BATCTPA**.***ILIBATTSCANSW (LTE eNodeB, 5G gNodeB)*** • **BATCTPA**.***ILIBATTSCTM (LTE eNodeB, 5G gNodeB)*** Modified parameters: • **DSP PMU**: Added support for querying the functions supported by the PMU. • **DSP BATTR**: Added support for querying intelligent lithium battery capacity verification test results.	3900 and 5900 series base stations (macro base stations)
Added the mutually exclusive relationship between PSU current limiting protection in case of AC circuit breaker capability insufficiency and iLiBattScan. For details, see **11.3.2.2 Mutually Exclusive Functions**.	None	3900 and 5900 series base stations (macro base stations)

Change Description	Parameter Change	Base Station Model
Added support for ICC power system management on the MAE. ● Added the mapping between ICC power cabinets and power system types. For details, see **4.3.3 Hardware**. ● Lifted the restriction that PSUs in an MTS power cabinet must be configured in an ascending order (starting from slot 1) and added support for configuring PSUs of the **SSU** type in MTS power cabinets. For details, see **4.4.1.1 Data Preparation**. ● Added parameter configurations related to ICC power cabinets. For details, see **4.4.1.1 Data Preparation**, **4.4.1.2 Using MML Commands**, and **4.6 Operation and Maintenance (Multimode Base Station)**.	Added the **PMUEXTINFO** MO and related parameters. Added the **DIESELGEN**.***DIESELGENMFR (5G gNodeB, LTE eNodeB)*** parameter.	● 3900 and 5900 series base stations (macro base stations) ● DBS3900 LampSite and DBS5900 LampSite
Added support for lithium battery software lock management when third-party power supply is used and support for configuring the lithium battery software lock timeout threshold when the PMU 15A is used. For details, see **13.3.3 Hardware** and **13.4.1.1 Data Preparation**.	Added the **IPWRM**.***LIBATTLOCKTIMEOUT (LTE eNodeB, 5G gNodeB)*** parameter.	3900 and 5900 series base stations (macro base stations)

Editorial Changes

Revised descriptions of support for lithium battery power backup and lithium battery software lock management when the PMU 15A or PMU 15A (Ver.B) is used. For details, see **4.3.3 Hardware**.

Revised other descriptions in this document.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in this document apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Applicable RAT

This document applies to GSM, UMTS, LTE FDD, LTE TDD, LTE NB-IoT, and NR. This feature works the same way with each of these RATs.

2.3 Features in This Document

The following table lists the mapping between features for different RATs and chapters/sections.

RAT	Feature ID	Feature Name	Chapter/Section
GSM	GBFD-510710	Intelligent Battery Management	**5 Intelligent Battery Management**
UMTS	WRFD-140220	Intelligent Battery Management	
LTE FDD	LOFD-001071	Intelligent Battery Management	
LTE TDD	TDLOFD-001071	Intelligent Battery Management	
LTE NB-IoT	MLOFD-001071	Intelligent Battery Management	
GSM	MRFD-221204	iLiBattScan (GSM)	**7 iLiBattScan**
UMTS	MRFD-221214	iLiBattScan (UMTS)	
LTE FDD	MRFD-221224	iLiBattScan (LTE FDD)	
LTE TDD	MRFD-221234	iLiBattScan (LTE TDD)	
NR	MRFD-221264	iLiBattScan (NR)	

3 General Principles

Each base station has a power supply system, which converts and distributes power and manages power-related alarms and batteries. This section describes the power supply system in power sources for a base station.

The power supply system consists of the power monitoring unit (PMU), power supply unit (PSU), boost distribution unit (BDU), and storage battery. The functions of the PMU, PSU, and BDU are as follows:

- The PMU manages the power supply, monitors power distribution, and reports alarms.
- The PSU converts voltages and monitors alarms related to module faults.
- The BDU supports output of boosted voltage and monitors alarms related to module faults.

Huawei AC-powered base stations support basic configuration, information query, and software management for the PMUs, PSUs, BDUs, and storage batteries. In addition, they can provide the following functions based on network requirements:

- **5 Intelligent Battery Management**
- **6 Automatic Battery Testing Management**
- Base station hierarchical power-off. For details, see *Green Site* in each single-RAT feature documentation package.
- **7 iLiBattScan**
- **8 Reporting of Loss of Power Supply Redundancy**
- **9 Diesel Generator Testing Management**
- **10 Intelligent Diesel Generator Management**
- **11 PSU Current Limiting Protection in Case of AC Circuit Breaker Capability Insufficiency**
- **12 Hybrid Use of Lead-Acid and Lithium Batteries**
- PV deployment for base stations. For details, see *Green Site* in each single-RAT feature documentation package.
- PMU support for associated temperature control of battery cabinets. For details, see *Green Site* in each single-RAT feature documentation package.
- **13 Lithium Battery Software Lock Management**

- **14 Optimized Access Policy on Power Monitoring Modules**
- Third-party green power access measurement and O&M management. For details, see *Green Site* in each single-RAT feature documentation package.
- Reference busbar voltage configuration. For details, see descriptions of the **Busbar Voltage Reference** parameter in **4.4.1.1 Data Preparation**.
- Staggering electricity usage. For details, see *Green Site* in each single-RAT feature documentation package.
- Precise backup power saving. For details, see *Green Site* in each single-RAT feature documentation package.

NOTE

In scenarios where base stations share one power system, for example, in separate-MPT scenarios, it is recommended that the power system be configured only on the NE of a more recent RAT. For example, in a UL co-BBU separate-MPT scenario, it is recommended that the PMU be configured only on the LTE side. If the power system is configured on both NEs, the configurations must be consistent. If the configurations are inconsistent, both NEs will report ALM-26271 Inter-System Monitoring Device Parameter Settings Conflict.

4 Basic Power Supply Management Functions

4.1 Principles

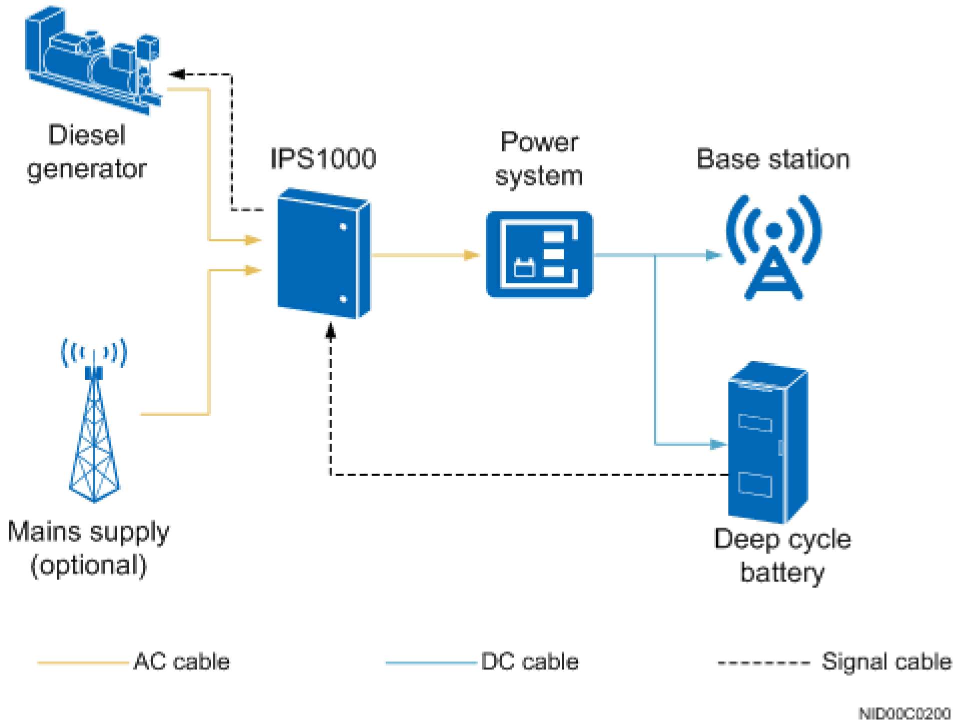
The power source for a base station may be the mains supply, diesel generator, solar sub-array, wind turbine generator, and batteries.

Huawei provides the following power sources:

- Mains supply
- Diesel generator
- Diesel generator and battery hybrid power sources
- Solar and wind hybrid power sources
- Solar and diesel generator hybrid power sources

These sources provide a reliable power supply even without a mains supply. **Figure 4-1**, **Figure 4-2**, **Figure 4-3**, and **Figure 4-4** show the base station power sources.

Figure 4-1 Diesel generator and battery hybrid power sources



NOTE

The intelligent-hybrid power system (IPS1000) provides various functions, such as battery monitoring management, power switchover, and power monitoring.

Figure 4-2 Dual-diesel generator power sources

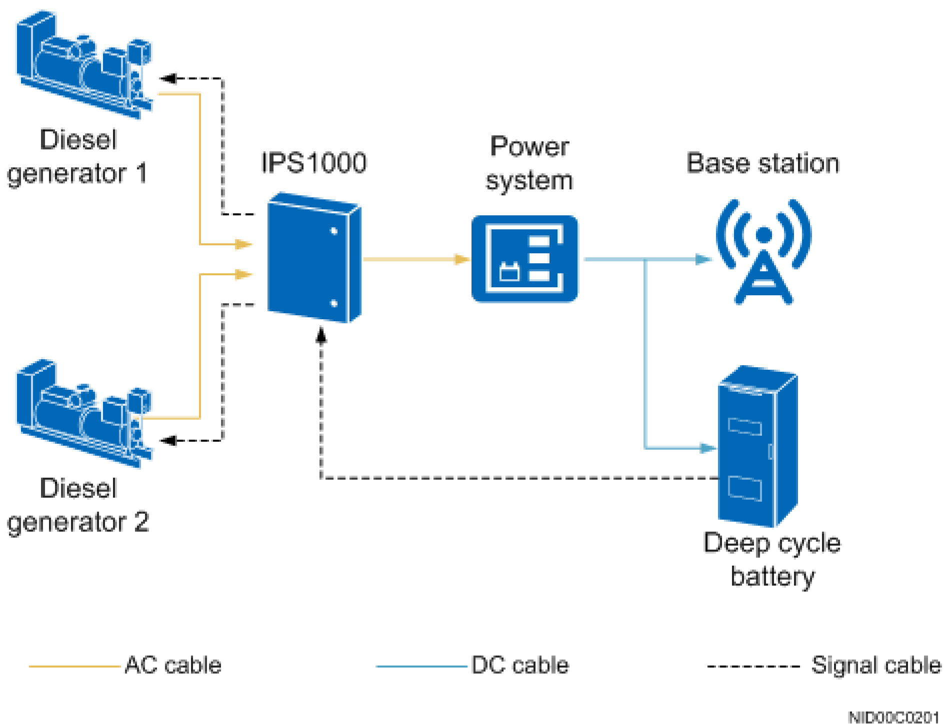


Figure 4-3 Solar and wind hybrid power sources

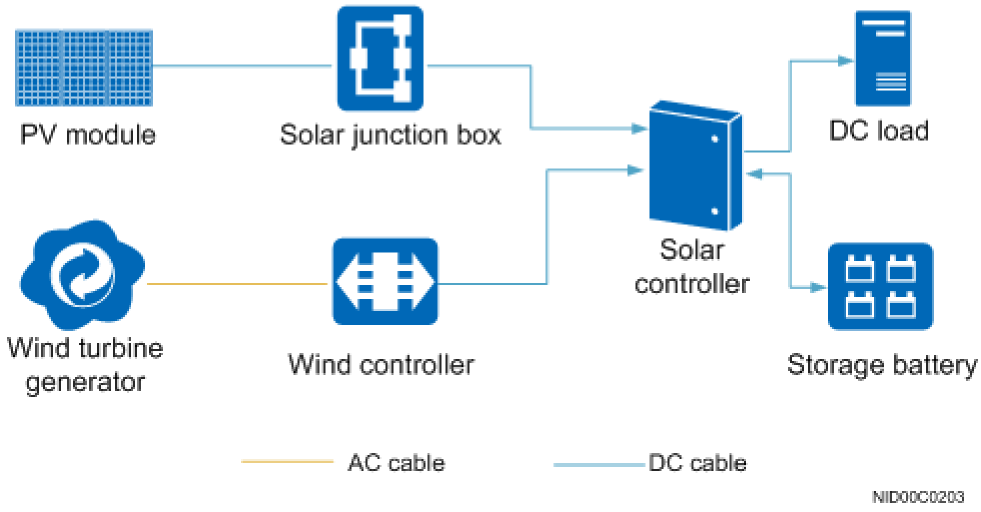
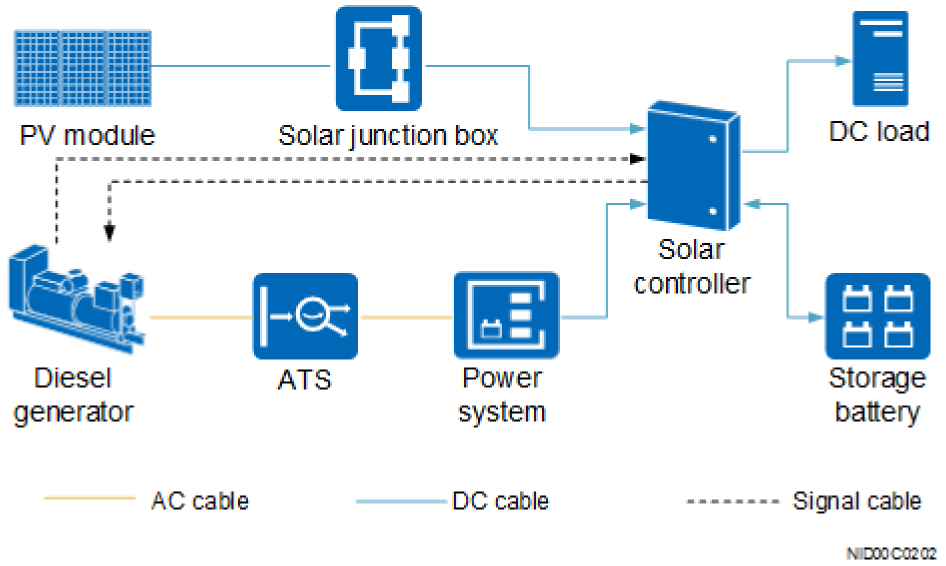


Figure 4-4 Solar and diesel generator hybrid power sources



NOTE

The automatic transfer switch (ATS) automatically starts and stops the diesel generator, switches AC power between the mains supply and the diesel generator, and provides surge protection.

4.2 Network Analysis

4.2.1 Benefits

None

4.2.2 Impacts

None

4.3 Requirements

4.3.1 Licenses

There are no license requirements for basic functions.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

None

4.3.2.2 Mutually Exclusive Functions

None

4.3.3 Hardware

Base Station Models

- 3900 and 5900 series base stations (macro base stations)
- DBS3900 LampSite and DBS5900 LampSite

Boards

The BBU that is directly connected to the PMU must be configured with the UPEU or UEIU.

RF Modules

For RRU remote monitoring, the RRU is required to provide a monitoring port or transmit monitoring information through a power cable. For details about whether an RRU supports a monitoring port, see section "Ports" in the corresponding RRU hardware description.

Power Supply

The PMU has multiple power system types, such as ETP. The power system type can be specified by the **PMU**.***PTYPE*** parameter. In the following tables, "√" indicates that the function is supported and "x" indicates that the function is not supported.

Table 4-1 Capabilities of ETP power supply systems to support the functions

PMU Hardware Model	Power Supply System Type	Intelligent Battery Management	Base Station Hierarchical Power-Off	Reporting of the Loss of Power Supply Redundancy Alarm	Lithium Battery Power Backup	PSU Current Limiting Protection in Case of AC Circuit Breaker Capability Insufficiency	Reference Busbar Voltage Configuration[1]	Coordinated Battery Cabinet Temperature Control	Lithium Battery Software Lock Management
PMU 11A	ETP	√	√	√	x	x	x	x	x
PMU 11B	ETP	√	√ The backup power capability for power distribution slots is supported. The PMU software version must be 219 or later.	√	x	√ (The PMU software version must be 200 or later.)	√ (The PMU software version must be 219 or later.)	x	x

PMU Hardware Model	Power Supply System Type	Intelligent Battery Management	Base Station Hierarchical Power-Off	Reporting of the Loss of Power Supply Redundancy Alarm	Lithium Battery Power Backup	PSU Current Limiting Protection in Case of AC Circuit Breaker Capability Insufficiency	Reference Busbar Voltage Configuration[1]	Coordinated Battery Cabinet Temperature Control	Lithium Battery Software Lock Management
PMU 15A/PMU 15A (Ver.B)	ETP	x	x	x	√ (When the hardware version is PMU.19, the PMU software version must be 166 or later. When the hardware version is PMU.40, all software versions can be used.)	x	x	x	√ (The setting of the lithium battery software lock timeout threshold is supported when the PMU software version is later than 2307 and the hardware version is PMU.40.)

PMU Hardware Model	Power Supply System Type	Intelligent Battery Management	Base Station Hierarchical Power-Off	Reporting of the Loss of Power Supply Redundancy Alarm	Lithium Battery Power Backup	PSU Current Limiting Protection in Case of AC Circuit Breaker Capability Insufficiency	Reference Busbar Voltage Configuration[1]	Coordinated Battery Cabinet Temperature Control	Lithium Battery Software Lock Management
PMU 16A	ETP	√	√ ● The non-backup power capability for power distribution slots is supported. ● The backup power capability for power distribution slots is supported. The PMU software version must be 177 or later.	√	√	√ (When the hardware version is PMU.29, the PMU software version must be 168 or later.)	x	x	x
OPM50M	ETP	√	x	x	√ (The PMU software version must be 129 or later.)	x	x	x	x

PMU Hardware Model	Power Supply System Type	Intelligent Battery Management	Base Station Hierarchical Power-Off	Reporting of the Loss of Power Supply Redundancy Alarm	Lithium Battery Power Backup	PSU Current Limiting Protection in Case of AC Circuit Breaker Capability Insufficiency	Reference Busbar Voltage Configuration[1]	Coordinated Battery Cabinet Temperature Control	Lithium Battery Software Lock Management
OPM30M	ETP	√	x	x	√ (The PMU software version must be 102 or later.)	x	x	x	x
OPM30A	ETP	√	x	x	√ (The PMU software version must be 102 or later.)	x	x	x	x

PMU Hardware Model	Power Supply System Type	Intelligent Battery Management	Base Station Hierarchical Power-Off	Reporting of the Loss of Power Supply Redundancy Alarm	Lithium Battery Power Backup	PSU Current Limiting Protection in Case of AC Circuit Breaker Capability Insufficiency	Reference Busbar Voltage Configuration[1]	Coordinated Battery Cabinet Temperature Control	Lithium Battery Software Lock Management
CCUC	ETP	√	√ ● When the hardware version is PMU.19, the PMU software version must be 219 or later. ● When the hardware version is PMU.22, the PMU software version must be 213 or later.	√	√ ● The PMU software version must be 168 or later. ● When the ESM-48150B1 or ESM-48100U2 is used, the PMU software version must be 2251 or later.	√ ● When the hardware version is PMU.19, the PMU software version must be 208 or later. ● When the hardware version is PMU.22, the PMU soft	√ ● When the hardware version is PMU.19, the PMU software version must be 219 or later. ● When the	x	√ (The PMU software version must be later than 2307.)

PMU Hardware Model	Power Supply System Type	Intelligent Battery Management	Base Station Hierarchical Power-Off	Reporting of the Loss of Power Supply Redundancy Alarm	Lithium Battery Power Backup	PSU Current Limiting Protection in Case of AC Circuit Breaker Capability Insufficiency	Reference Busbar Voltage Configuration[1]	Coordinated Battery Cabinet Temperature Control	Lithium Battery Software Lock Management
						ware version must be 203 or later.	hardware version is PMU.22, the PMU software version must be 213 or later.		
PMU 11C	ETP	√	√	√	√	√	√	√	√ (The PMU software version must be later than 2309.)

Table 4-2 Capabilities of other power supply systems to support the functions

PMU Hardware Model	Power Supply System Type	Intelligent Battery Management	Base Station Hierarchical Power-Off	Reporting of the Loss of Power Supply Redundancy Alarm	Lithium Battery Power Backup	PSU Current Limiting Protection in Case of AC Circuit Breaker Capability Insufficiency	Reference Busbar Voltage Configuration
PMU 01B	EPS4890	x	x	√	x	x	x
PMU 11A	EPS4890	√	√	√	x	x	x
PMU 11B	EPS4890	√	√	√	x	√	x
SC48200	SC48200	x	x	x	x	x	x
PMU 11A	EPS4815	√	x	√	√ (The PMU software version must be 137 or later.)	x	x
PMU 11B	EPS4815	√	x	√	√ (The PMU software version must be 137 or later.)	x	x
SMU02C (02312MML-009)	SMU	x	√	x	√	x	√

Before upgrading the power system to support 57.0 V, ensure that the reference busbar output voltage can be 57.0 V. The reference busbar output voltage is related to the PSU model and storage batteries, as listed in the following table. If the PSUs and storage batteries do not meet the requirements, replace them according to the following table. Then, set the **Busbar Voltage Reference** parameter to **57.0V** for the PMU and reset the PMU for the configuration to take effect.

Table 4-3 Reference busbar output voltage

PSU Model	Storage Battery	Reference Busbar Output Voltage
R4875G5 (supporting 57 V output)	Lead-acid storage batteries	53.5 V
	Hybrid use of lead-acid and lithium batteries	53.5 V
	Lithium batteries[1]	57.0 V and 53.5 V
	No battery	57.0 V and 53.5 V
R4875G1/R4875G2 (not supporting 57 V output)	Lead-acid storage batteries	53.5 V
	Hybrid use of lead-acid and lithium batteries	53.5 V
	Lithium batteries	53.5 V
	No battery	53.5 V

NOTE

[1]: In the lithium battery backup scenario, if the reference busbar voltage has been set to 57.0 V but a PMU software version that does not support the reference busbar voltage configuration is activated, the value of the **PMU**.***BVR (LTE eNodeB, 5G gNodeB)*** parameter specifying the reference busbar voltage must be changed to **DEFAULT**. Otherwise, the lithium batteries may fail to provide backup power.

Table 4-4 Mapping between the power system and lithium battery

Power System	Lithium Battery
APM5930	ESM-48100B1 ESM-48150B1 ESM-48100U2
APM30H (Ver.B to Ver.E)	ESM-48100B1 ESM-48150B1 ESM-48100U2
OPM200/OPM200 (Ver.B)	IBBS50L/IBBS50L-F-02
OPM50M	IBBS20L
APM5950H	ESM-48100B1 ESM-48150B1 ESM-48100U2

Power System	Lithium Battery
APM5950H-L	ESM-48100B1 ESM-48150B1 ESM-48100U2

PMU Software Versions

- The PMU software version must be SRAN7.0 or later.
- If the power supply system type is EPS4815, the PMU software version must be SRAN9.0 or later.

Mapping Between Physical Cabinets and PTYPE Values

Before configuring power modules for a base station, learn the mapping between physical cabinets and power supply system types (specified by the **PMU**.***PTYPE*** parameter). The following tables list the mapping between physical cabinets and PTYPE values.

Table 4-5 Mapping between physical cabinets and power supply system types (1)

Cabinet Name	Power Module Hardware Model	PTYPE
APM30H (Ver.C)	PMU 01B	APM30
APM30H (Ver.D) APM30H (Ver.D_A1) APM30H (Ver.D_A2) APM30H (Ver.D_B)	PMU 11A	APM30
APM30H (Ver.E)	CCUB/CCUC	APM30
BTS3900 (Ver.C)	PMU 01B	EPS4890
BTS3900 (Ver.D) BTS3900 (Ver.D_A) BTS3900 (Ver.D_B)	PMU 11A	EPS4890
BTS3900L (Ver.D)	PMU 11A	EPS4890

Table 4-6 Mapping between physical cabinets and power supply system types (2)

Cabinet Name	Power Module Hardware Model	PTYPE
PS4890	PMU 11A	EPS4890
OMB (Ver.C)	PMU 11A/PMU 11B	EPS4815

Cabinet Name	Power Module Hardware Model	PTYPE
IMB05	PMU 11B	EPS4815
SC48200	SC48200	SC48200
SC48200	ICC100-N5	SC48200
SC4850	SC48200	SC48200

Table 4-7 Mapping between physical cabinets and power supply system types (3)

Cabinet Name	Power Module Hardware Model	PTYPE
BTS3900AL (Ver.A)	PMU 11A	ETP
TP48600A-H17B1	PMU 11A	ETP
IMS06	PMU 11A	ETP
MRE1000	CCUB	ETP
APM5930(AC) (Ver.A)	CCUC	ETP
BBC5200D-L/BBC5600D/BBC5600A	PMU 15A/PMU 15A (Ver.B)	ETP
IBC10	PMU 11B	ETP
IBC10 (Ver.B)	PMU 11B	ETP
APM5950H/APM5950H-L	PMU 11C	ETP
MTS9304A-HX10A1 (01075308) MTS9514A-AM2003 (01075875)	SMU02C (02312MML-009)	SMU
ICC30-N1/ICC200-N1-C12/ICC330-D1/ICC330-H1-C6/ICC330-H1-C8/ICC330-HA1-C7/ICC330-HA1-C10/ICC330-HA1-C11/ICC330-HD1-C6/ICC350-H1-C5/ICC360-HA1-C1/ICC360-HA1-C2/ICC720-HA1-C2/ICC800-A1-C2	SMU02C (02312MML-009)	SMU

Mapping Between Physical Power Modules and PTYPE Values

If the site is configured with any of the following power modules, the corresponding power supply system type (specified by the **PMU**.***PTYPE*** parameter) also needs to be configured. The following table lists the mapping between physical power modules and PTYPE values.

Table 4-8 Mapping between physical power modules and power supply system types

Power Module Name	Power Module Hardware Model	PTYPE
OPM50M	OPM50M	ETP
OPM50M (Ver.B)[1]	OPM50M	ETP
OPM30A	OPM30A	ETP
OPM200/OPM200 (Ver.B)	PMU16A	ETP
EPU02D/EPU02D-02	PMU 12A	EPU
EPU02B/EPU02S	PMU 12A	EPU
EPU02S[2]	SDU60-02	EPU
EPU02B[2]	BDU70-03	EPU
EPU02S-02[2]	SDU60-02/SDU30-01	EPU

NOTE

[1] The BBU3910C can be powered by the mains power, diesel generator, and solar power equipment. Because the power consumption of the BBU3910C is low, it is mainly powered by an ICC100-N5. For details about the load output circuit breakers on the ICC100-N5, see position 8 in section "Power Distribution" of *ICC100-N5 Solar Controller User Manual*. For details about the power cable connections on the BBU3910C side, see the section "Installing a Power Cable" in *BBU3910C Installation Guide*. For details about the requirements for upper-level circuit breakers, see the section "Installation Environment" in *BBU3910C Installation Guide*.

[2] If there is no PMU 12A in the voltage boosting power distribution box, a virtual PMU can be implemented based on the SDU60-02/BDU70-03 and the monitoring function can be realized using RRUs. The type and communication address of the PMU are set to EPU and 16, respectively. After the virtual PMU is implemented, the electronic label of the module is queried through the **PMU** MO and the return value of the **BDU** MO is empty.

4.3.4 Others

N/A

4.4 Operation and Maintenance (eGBTS/NodeB/eNodeB/gNodeB)

A 3900 or 5900 series base station must be configured with a PMU and PSUs to provide basic power supply, power distribution, and power management functions.

The storage batteries and diesel generators can be configured based on site requirements.

4.4.1 Activation and Optimization

4.4.1.1 Data Preparation

Scenario 1: PMU Configuration

Table 4-9 describes the parameter settings for PMU configuration.

Table 4-9 Parameter settings for PMU configuration

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	**PMU**.***CN (LTE eNodeB, 5G gNodeB)***	Set this parameter based on the actual situation. SMUs can only be configured in virtual cabinets.
Subrack No.	**PMU**.***SRN (LTE eNodeB, 5G gNodeB)***	Generally, the default value **7** is used. For the voltage boosting power distribution subrack, the value range is **17–18**.
Slot No.	**PMU**.***SN (LTE eNodeB, 5G gNodeB)***	Generally, the default value **0** is used.
Power System Type	**PMU**.***PTYPE (LTE eNodeB, 5G gNodeB)***	• The power supply system type must match the physical cabinet. For the mapping between cabinet types and power supply system types, see **Mapping Between Physical Cabinets and PTYPE Values**. • When **BATTERY**.***BTYPE*** is set to **VRLA_BAT** or **VRLA_BAT_MIX_LI_BAT**, **PMU**.***BVR*** must be set to **53.5V** or **DEFAULT** for the PMU that manages the batteries.
Manager Cabinet No.	**PMU**.***MCN (LTE eNodeB, 5G gNodeB)***	The manager of a PMU is usually a BBU or RRU. Set this parameter to the cabinet number of the BBU or RRU. An SMU must be managed by either a BBU or an RRU.
Manager Subrack No.	**PMU**.***MSRN (LTE eNodeB, 5G gNodeB)***	The manager of a PMU is usually a BBU or RRU. Set this parameter to the subrack number of the BBU or RRU. The EPU02B/EPU02S/EPU02S-02 can be remotely monitored through the BDU/SDU power cable that supplies power to the RRU. The subrack number is that of the RRU that receives power from the BDU/SDU. An SMU must be managed by either a BBU or an RRU.

Parameter Name	Parameter ID	Setting Notes
Manager Port No.	**PMU**.***MPN (LTE eNodeB, 5G gNodeB)***	● If the PMU is connected to the MON1 port on the UPEU, set this parameter to **1**. ● If the PMU is connected to the MON0 port on the UPEU, set this parameter to **0**. ● If the PMU is connected to an RRU, set this parameter to **0**.
Address	**PMU**.***ADDR (LTE eNodeB, 5G gNodeB)***	The PMU supports DIP switch values of **3** (default value), **4**, and **9**[1]. To modify the communication address, you must modify both the parameter value and the value of the DIP switch.
AC Voltage Alarm Lower Threshold	**PMU**.***ACVLTHD (LTE eNodeB, 5G gNodeB)***	Set this parameter according to the power supply specifications. The value **180** is recommended.
AC Voltage Alarm Upper Threshold	**PMU**.***ACVUTHD (LTE eNodeB, 5G gNodeB)***	Set this parameter according to the power supply specifications. The value **280** is recommended.
DC Voltage Alarm Lower Threshold	**PMU**.***DCVLTHD (LTE eNodeB, 5G gNodeB)***	If lead-acid batteries are used to provide power backup: ● **470** is recommended for the SC48200 PMU. ● **450** is recommended for other PMUs. If lithium batteries are used to provide power backup[2]: ● **500** is recommended for a power supply system using the IBBS20L to provide power backup. ● **477** is recommended for a power supply system using the ESM-48100B1/ ESM-48150B1/ESM-48100U2 to provide power backup. ● **461** is recommended when the APM30H (Ver.B)/APM30H (Ver.C)/APM30H (Ver.D)/ APM30H (Ver.E) cabinet uses lithium batteries for power backup. ● When **PMU**.***PTYPE*** is set to **SMU**, **PMU**.***DCVLTHD*** must be set to a value greater than or equal to **360**.

Parameter Name	Parameter ID	Setting Notes
DC Voltage Alarm Upper Threshold	**PMU**.***DCVUTHD (LTE eNodeB, 5G gNodeB)***	**585** is recommended for the SC48200 PMU. **580** is recommended for other PMUs. ● This parameter can be set to a value ranging from **530** to **600** when **PMU**.***PTYPE*** is set to **SMU**. ● This parameter can be set to a value ranging from **580** to **600** when **PMU**.***PTYPE*** is set to a value other than **SMU**.
Special Analog Alarm Flag	**PMU**.***SAAF (LTE eNodeB, 5G gNodeB)***	Set this parameter based on the actual situation. You are not advised to configure any sensor on the PMU. Therefore, the **PMU**.***SAAF*** parameter is generally disabled.
Special Boolean Alarm Flag	**PMU**.***SBAF (LTE eNodeB, 5G gNodeB)***	Set this parameter based on the actual situation.
Cabinet Temperature Alarm Lower Threshold	**PMU**.***ATLTHD (LTE eNodeB, 5G gNodeB)***	**0** is recommended for non-SMU PMUs, while **-20** is recommended for SMUs.
Cabinet Temperature Alarm Upper Threshold	**PMU**.***ATUTHD (LTE eNodeB, 5G gNodeB)***	**50** is recommended for non-SMU PMUs, while **55** is recommended for SMUs.
Cabinet Humidity Alarm Lower Threshold	**PMU**.***AHLTHD (LTE eNodeB, 5G gNodeB)***	**10** is recommended for non-SMU PMUs, while **5** is recommended for SMUs.
Cabinet Humidity Alarm Upper Threshold	**PMU**.***AHUTHD (LTE eNodeB, 5G gNodeB)***	**80** is recommended for non-SMU PMUs, while **95** is recommended for SMUs.
HVDU Number	**PMU**.***HVDUNUMBER***	Set this parameter based on the actual situation. An incorrect configuration will affect the reporting of the alarm related to the safety of 225–400 V DC power distribution.
HVDC Insulation Resistance Alarm Threshold	**PMU**.***HVDCIRAT***	The insulation resistance is that between the positive/negative bus bar and the ground. The parameter retains the default value. If the load resistance value is less than the threshold, contact Huawei technical support to check the threshold.
HVDC Output Voltage	**PMU**.***HVDCOUTVOLTAGE***	Set this parameter based on the actual situation. An incorrect parameter value may lead to overvoltage protection or even damage.

Parameter Name	Parameter ID	Setting Notes
Busbar Voltage Reference	**PMU**.***BVR***	● If this parameter is set to **53.5V**, the actual busbar voltage dynamically changes from –43.2 V to –58 V based on the battery charge and discharge status. ● If this parameter is set to **57.0V**, the actual busbar voltage changes from –55 V to –57 V. ● This parameter cannot be set to **AUTO** for SMUs.

NOTE

[1]: Set the PMU communication address as follows:

- Set the communication address of a single PMU to **3**.
- When PMUs are configured:
  - If two PMUs are configured on different RS485 buses, set the communication addresses to **3**.
  - If two PMUs are cascaded on one RS485 bus, set the communication address of the upper-level PMU to **3** and set the communication address of the lower-level PMU to **4**.
- There is no DIP switch for the PMU of the OPM50M. The BBU or RRU monitors the cascaded position of the OPM50M to automatically generate a communication address for the PMU.
  - If the OPM50M is monitored by a BBU3910A, there can only be a maximum of three-level OPM50M cascading on the RS485 bus. Set the communication address of the upper-level PMU to **3**, set that of the medium-level PMU to **4**, and set that of the lower-level PMU to **9**.
  - If the OPM50M is monitored by an RRU, cascading is not supported, and the communication address is **3** by default.
- The OPM30M does not support cascading connection, and the communication address is **3** by default.
- When PMUs are configured in a BDU:
  - If two PMUs are configured on different RS485 buses, set the communication addresses to **16**.
  - If two PMUs are cascaded on one RS485 bus, set the communication address of the upper-level PMU to **16** and set the communication address of the lower-level PMU to **17**.
  - When the PMU 12A connects to the MON0 or MON1 bus through an RS485 port, two PMU 12A modules must be cascaded before being connected to the bus to avoid PMU address conflict.
  - If the PMU is remotely monitored through the RRU power cable, set the **Manager Port No.** parameter to **0** and the communication address to **16**.
  - Set the communication address of the PMU 15A/PMU 15A (Ver.B) to **4**.

[2]: The end-of-discharge voltage of the lithium battery is higher than that of the lead-acid battery. If the **PMU**.***DCVLTHD*** parameter is set to the value recommended for the lead-acid battery in a lithium battery backup scenario, the base station cannot report ALM-25621 Power Supply DC Output Out of Range properly.

Table 4-10 Parameter settings for PMU extended information

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	**PMUEXTINFO**.***CN***	Set this parameter based on the actual situation.
Subrack No.	**PMUEXTINFO**.***SRN***	Generally, it is set to **7**.
Slot No.	**PMUEXTINFO**.***SN***	Generally, it is set to **0**.
Number of Diesel Generators	**PMUEXTINFO**.***DIESELGENNUM***	Set this parameter based on the network plan. The value of this parameter cannot exceed 1.
Bat Depth of Discharge for DG Startup	**PMUEXTINFO**.***BATDODFORDGSTARTUP***	Set this parameter based on the network plan. The value **60** is recommended.
Diesel Generator Minimum Charge Time	**PMUEXTINFO**.***MINCHGTIME***	The value **0** is recommended.
Diesel Generator Maximum Charge Time	**PMUEXTINFO**.***MAXCHGTIME***	The value **1440** is recommended.
Diesel Generator On or Off Judge Time	**PMUEXTINFO**.***ONOFFJUDGETIME***	The value **4** is recommended.
Diesel Generator Soft Start Time	**PMUEXTINFO**.***SOFTSTARTTIME***	The value **1** is recommended.

NOTE

Parameters in the **PMUEXTINFO** MO must be configured when the PMU type is SMU and diesel-grid hybrid power supply is used.

Scenario 2: PSU Configuration

Table 4-11 describes the parameter settings for PSU configuration.

Table 4-11 Parameter settings for PSU configuration

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	**PSU**.***CN (LTE eNodeB, 5G gNodeB)***	Set this parameter based on the actual situation.
Subrack No.	**PSU**.***SRN (LTE eNodeB, 5G gNodeB)***	Generally, the default value is used.

Parameter Name	Parameter ID	Setting Notes
Slot No.	**PSU**.***SN (LTE eNodeB, 5G gNodeB)***	• Configure the PSU in any of slots 1 to 3 in the APM30H (Ver.C). • Configure the PSU in any of slots 1 to 5 in the APM30H (Ver.D) or IBC10 AC cabinet. • Configure the PSU in any of slots 1 to 5 in the APM30H (Ver.E) AC cabinet. • Configure the PSU in slot 1 or 2 in the OMB (Ver.C). • Configure the PSU in any of slots 1 to 7 in the BTS3900AL (Ver.A) or TP48600A-H17B1. • Configure the PSU in any of slots 1 to 3 in the AC BTS3900 (Ver.C), BTS3900 (Ver.D), or PS4890. • The PSU cannot be configured when the power supply system type is SC48200 or SC4850. • Configure the PSU in any of slots 1 to 3 in an MTS or ICC power cabinet.
PSU Type	**PSU**.***PSUTYPE***	Set this parameter based on the network plan.

Scenario 3: BDU Configuration

Table 4-12 describes the parameter settings for BDU configuration.

Table 4-12 Parameter settings for BDU configuration

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	**BDU**.***CN (LTE eNodeB, 5G gNodeB)***	Set this parameter based on the actual situation.
Subrack No.	**BDU**.***SRN (LTE eNodeB, 5G gNodeB)***	The setting of this parameter must be consistent with the setting of the subrack number of the PMU.
Slot No.	**BDU**.***SN (LTE eNodeB, 5G gNodeB)***	• Configure the BDU in any of slots 6 to 9 in the APM30H (Ver.E) AC cabinet. • Configure the BDU in slot 1 or 2 in the voltage boosting power distribution box. • Configure the BDU in any of slots 6 to 10 in the APM5930 (Ver.A) AC cabinet.

Parameter Name	Parameter ID	Setting Notes
BDU Type	**BDU**.***BDUTYPE (LTE eNodeB, 5G gNodeB)***	• If this parameter is set to **DEFAULT**, the system does not check the consistency between the configuration type and the physical type of the device. The device runs according to the actual type. • If this parameter is set to **SDU**, **HSU**, or **BDU**, the system checks whether the type of the SDU, HSU or BDU in use is consistent with the configuration.
Voltage Adjustment Mode	**BDU**.***VOLADJMODE***	• The value **ADAPTIVE** indicates that the system adaptively adjusts the output voltage based on the actual load of the system. • The value **FIXED** indicates that the output voltage is always 57 V. • The default value is **ADAPTIVE**.

NOTE

BDUs cannot be configured in MTS or ICC power cabinets.

Scenario 4: Storage Battery Configuration

NOTE

- If the EPS4815 PMU is configured, no storage battery can be configured.
- If the SC48200 PMU is configured, storage batteries are configured automatically by default.
- If other PMUs are configured, storage batteries are configured as required.
- The configured battery parameters must be the same as the actual values. If the capacity and charge current limiting coefficient are different from the actual values, the charge current does not match the actual capacity of the capacitor, which affects the battery lifespan and even damages the battery.
- The OPM50M and OPM30M can be configured with lithium batteries.
- If batteries are not configured after they are installed, the batteries may not be charged as expected, affecting the battery power backup.

Table 4-13 describes the parameter settings for storage battery configuration.

Table 4-13 Parameter settings for storage battery configuration

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	**BATTERY**.***CN (LTE eNodeB, 5G gNodeB)***	Set this parameter based on the actual situation.
Subrack No.	**BATTERY**.***SRN (LTE eNodeB, 5G gNodeB)***	Generally, the default value is used.

Parameter Name	Parameter ID	Setting Notes
Slot No.	**BATTERY**.***SN (LTE eNodeB, 5G gNodeB)***	The default value **0** is recommended.
Installation Type	**BATTERY**.***INSTALLTYPE (LTE eNodeB, 5G gNodeB)***	The installation type of the storage battery for the SC48200 PMU must be **OUTER**. Set this parameter as required for other types of PMU.
Battery Type	**BATTERY**.***BTYPE (LTE eNodeB, 5G gNodeB)***	Set this parameter based on the storage battery type.
Number Of Lithium Batteries	**BATTERY**.***LBN (LTE eNodeB, 5G gNodeB)***	This parameter is valid only when the **BTYPE** parameter is set to **LI_BAT** or **VRLA_BAT_MIX_LI_BAT**.
Number Of Battery Groups	**BATTERY**.***BN (LTE eNodeB, 5G gNodeB)***	This parameter is valid only when the **BTYPE** parameter is set to **VRLA_BAT**. The value **1** is recommended.
Lead-acid Battery 1 Capacity	**BATTERY**.***BC1 (LTE eNodeB, 5G gNodeB)***	Set these parameters based on the actual situation. Incorrect settings may affect the lifetime of the batteries. For the requirements on the configuration, see the product manual delivered with the storage batteries.
Lead-acid Battery 2 Capacity	**BATTERY**.***BC2 (LTE eNodeB, 5G gNodeB)***	
Boost-Charging Duration	**BATTERY**.***BCD (LTE eNodeB, 5G gNodeB)***	For the specific configuration, see the product manual delivered with the storage batteries.
Boost-Charging Voltage	**BATTERY**.***BCV (LTE eNodeB, 5G gNodeB)***	**565** is recommended
Float-Charging Voltage	**BATTERY**.***FCV (LTE eNodeB, 5G gNodeB)***	**535** is recommended.
Battery Current Limiting Coefficient	**BATTERY**.***BCLC (LTE eNodeB, 5G gNodeB)***	• The value **15** is recommended. • This parameter can be set (within the range from **5** to **25**) when the battery type is lithium battery and the APM5930/APM5950H/APM5950H-L/OPM30M/OPM40M/OPM120/OPM200/OPM200 (Ver.B) is used. • The parameter value ranges from **5** to **100** only when the battery type is lithium battery and the PMU type is SMU. The parameter can be set to up to **100** for 100 Ah lithium batteries, up to **68** for 150 Ah lithium batteries, and up to **50** for 200 Ah lithium batteries. If the configured value exceeds the capability of the lithium batteries, the configuration does not take effect.
Temperature Compensation Coefficient	**BATTERY**.***TCC (LTE eNodeB, 5G gNodeB)***	**80** is recommended.

Parameter Name	Parameter ID	Setting Notes
Temperature Alarm Lower Threshold	**BATTERY**.*TLTHD (LTE eNodeB, 5G gNodeB)*	**–19** is recommended.
Temperature Alarm Upper Threshold	**BATTERY**.*TUTHD (LTE eNodeB, 5G gNodeB)*	The value **50** is recommended.
Shutdown Voltage	**BATTERY**.*SDV*	If lead-acid batteries are used to provide power backup: ● **460** is recommended for the SC48200 PMU. ● **430** is recommended for other PMUs. If lithium batteries are used to provide power backup[1]: ● **482** is recommended for a power supply system using the IBBS20L to provide power backup. ● **462** is recommended for a power supply system using the ESM-48100B1/ESM-48150B1/ESM-48100U2 to provide power backup. ● **430** is recommended for the APM30H (Ver.B)/APM30H (Ver.C)/APM30H (Ver.D)/APM30H (Ver.E) cabinet using lithium batteries for power backup. ● When an MTS or ICC power cabinet uses lithium batteries for power backup, this parameter can be set to a value ranging from **360** to **556**. The value **432** is recommended.

NOTE

[1]: The end-of-discharge voltage of the lithium battery is higher than that of the lead-acid battery. If the default value for the lead-acid battery is used, discharge undervoltage protection will be performed on the lithium battery first. Then, power-off under low voltage will not take effect when it is enabled.

Scenario 5: Diesel Generator Configuration

Table 4-14 describes parameter settings for diesel generator configuration.

Table 4-14 Parameter settings for diesel generator configuration

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	**DIESELGEN**.*CN (LTE eNodeB, 5G gNodeB)*	Set this parameter based on the actual situation.

Parameter Name	Parameter ID	Setting Notes
Subrack No.	**DIESELGEN**.***SRN (LTE eNodeB, 5G gNodeB)***	Generally, the default value is used.
Slot No.	**DIESELGEN**.***SN (LTE eNodeB, 5G gNodeB)***	Generally, it is set to **0**.
Rated Power	**DIESELGEN**.***POWER (LTE eNodeB, 5G gNodeB)***	Set this parameter based on the rated output power printed on the name plate of the diesel generator. The value **125** is recommended.
Diesel Generator Manufacturer	**DIESELGEN**.***DIESELGENMFR***	Set this parameter based on the actual situation.
Diesel Generator Brand	**DIESELGEN**.***DIESELGENBRAND***	Set this parameter based on the actual situation.
Installation Date	**DIESELGEN**.***INSTDATE***	Set this parameter based on the network plan.
Output AC Phase Type	**DIESELGEN**.***OUTPUTACPT***	The value **THREE_PHASE_FOUR_WIRE** is recommended.
Rated Phase Voltage	**DIESELGEN**.***RATEDPHVOLT***	The value **230** is recommended.
Rated Frequency	**DIESELGEN**.***RATEDFREQ***	The value **50** is recommended.
Max Output Power Ratio	**DIESELGEN**.***MAXOUTPUTPWRRATIO***	The value **800** is recommended.

4.4.1.2 Using MML Commands

Before using MML commands, refer to **4.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

4.4.1.2.1 Activation Command Examples

NOTE

The following scenarios can exist simultaneously. If they do, MOs for all these scenarios must be configured.

Scenario 1: PMU Configuration

Step 1 Run the **ADD PMU** command to add a PMU. In this step, set the parameters in **Table 4-9**.

Step 2 Run the **ADD PMUEXTINFO** command to add the extended information of the PMU. (This command is required only when the PMU type is SMU and diesel-grid hybrid power supply is used.)

----End

NOTE

- The **PMU**.***MCN***, **PMU**.***MSRN***, **PMU**.***MPN***, and **PMU**.***ADDR*** parameters must be correctly specified to ensure proper communication between the PMU and BBU.
- **PTYPE** must be correctly specified. Then, set the **BTYPE**, **BCLC**, and **BC1** or **BC2** parameters based on the actual situation. Incorrectly setting the **BC1** or **BC2** parameter reduces the lifetime of the storage batteries.
- SMUs can only be configured in virtual cabinets. Before adding an SMU, run the **ADD CABINET** command with the **CABINET**.***TYPE*** parameter set to **VIRTUAL**.
- When the PMU is powered on again or reset, it configures the power system using its default parameter settings and then updates the parameter settings to match those on the BBU after being connected to the BBU.

Scenario 2: PSU Configuration

Step 1 Run the **ADD PSU** command to add PSUs. In this step, set the parameters in **Table 4-11**.

----End

NOTE

Removing, replacing, or reinstalling an existing PSU in an MTS power cabinet will trigger a communication failure alarm.

Scenario 3: BDU Configuration in a Voltage Boosting Power Distribution Box

Step 1 Run the **ADD PMU** command to add a PMU.

Step 2 Run the **ADD BDU** command to add a BDU. In this step, set the parameters in **Table 4-12**.

----End

NOTE

- The **PMU**.***MCN***, **PMU**.***MSRN***, **PMU**.***MPN***, and **PMU**.***ADDR*** parameters must be correctly specified to ensure proper communication between the PMU and BBU.
- **PTYPE** must be correctly specified. After setting **PTYPE**, configure the BDU based on site conditions.

Scenario 4: BDU/HSU Configuration (in an APM30H or MRE1000 Cabinet)

Step 1 Run the **ADD BDU** command to add a BDU. In this step, set the parameters in **Table 4-12**.

----End

Scenario 5: Storage Battery Configuration

Step 1 Run the **ADD BATTERY** command. In this step, set the parameters in **Table 4-13**.

----End

Scenario 6: Diesel Generator Configuration

Step 1 Run the **ADD DIESELGEN** to add a diesel generator. In this step, set the parameters in **Table 4-14**. For the PMU type SC48200, two diesel generators can be added. For the PMU type SMU, only one diesel generator can be added. Configuring one or two diesel generators follows the same method and procedure. The diesel generator test is supported only when the SC48200 is used. Steps 2 and 3 can be performed as required.

Step 2 Run the **STR DIESELGENTST** command to start the diesel generator test on a PMU.

Step 3 Run the **STP DIESELGENTST** command to stop the test.

----End

Scenario 7: Closing the Circuit Breaker in a 225–400 V DC Power Distribution Unit

Step 1 Run the **CLR HVDCOUTPUTPROTECT** command to close the circuit breaker in a 225–400 V DC power distribution unit.

----End

Scenario 8: Querying the Status of a 225–400 V DC Power Distribution Unit

Step 1 Run the **DSP HVDCSTATE** command to query the status of a 225–400 V DC power distribution unit.

----End

4.4.1.2.2 Deactivation Command Examples

N/A

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

Method 1: Run the **DSP BRD** command to query the availability status of each board in the power system.

Method 2: Check whether power devices such as the PMU, PSU, BDU, and PDB are normal on the MAE-Access or Web LMT device panel.

NOTE

The power devices in the PMU subrack, such as the PSU, BDU, and PDB, are managed by the PMU. The alarm status of these devices is displayed under the PMU.

In scenarios where batteries are configured, the **DSP BATTERY** command can be executed to query the overall dynamic information of all batteries. Alternatively, the **DSP BATTERYITEM** command can be executed to query the dynamic information of a specific battery by setting the **BATTERY**.***BATNO (LTE eNodeB, 5G gNodeB)*** parameter.

NOTE

- The **DSP BATTERYITEM** command can only be used to monitor wireless power devices, including the APM5930 (excluding the PMU.22 hardware version), APM5950, OPM200, and PMU 15A (Ver.B).
- The **DSP BATTERYITEM** command can only be used to query the dynamic information of lithium batteries.

4.5 Operation and Maintenance (GBTS)

A 3900 or 5900 series base station must be configured with a PMU and PSUs to provide basic power supply, power distribution, and power management functions.

The storage batteries and diesel generators can be configured based on site requirements.

4.5.1 Activation and Optimization

4.5.1.1 Data Preparation

Scenario 1: PMU Configuration

Table 4-15 describes the parameter settings for PMU configuration.

Table 4-15 Parameter settings for PMU configuration

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	**BTSBRD**.***CN***	Set this parameter based on the actual situation.
Subrack No.	**BTSBRD**.***SRN***	Generally, the default value is used.
Slot No.	**BTSBRD**.***SN***	Generally, the default value is used.
Board Type	**BTSBRD**.***BT***	Set this parameter to **PMU**.
Power Type	**BTSAPMUBP**.***PTYPE***	The power supply system type must match the physical cabinet.

Parameter Name	Parameter ID	Setting Notes
Special Analog Alarm Flag	**BTSAPMUBP**.***SAAF***	● You are not advised to configure any sensor on the PMU. Therefore, all sensors are generally disabled. That is, the parameter is set to **11111**. ● Battery temperature sensor 1 is enabled when the power supply system is PS4890 or SC48200 or when storage batteries are configured. It is disabled by default when any other power supply system is configured, such as the ETP.
Special Boolean Alarm Flag	**BTSAPMUBP**.***SBAF***	It is set to the disabled state for the water sensor and smoke sensor.
AC Voltage Alarm Lower Threshold	**BTSAPMUBP**.***ACVLTHD***	Set this parameter according to the power supply specifications. The value **180** is recommended.
AC Voltage Alarm Upper Threshold	**BTSAPMUBP**.***ACVUTHD***	Set this parameter according to the power supply specifications. The value **280** is recommended.
DC Voltage Alarm Upper Threshold	**BTSAPMUBP**.***DCVUTHD***	**585** is recommended for the SC48200 PMU. **580** is recommended for other power supply systems. ● This parameter can be set to a value ranging from **530** to **600** when **PMU**.***PTYPE*** is set to **SMU**. ● This parameter can be set to a value ranging from **580** to **600** when **PMU**.***PTYPE*** is set to a value other than **SMU**.
DC Voltage Alarm Lower Threshold	**BTSAPMUBP**.***DCVLTHD***	If lead-acid batteries are used to provide power backup: ● **470** is recommended for the SC48200 PMU. ● **450** is recommended for other power supply systems. If lithium batteries are used to provide power backup[2]: ● **500** is recommended for a power supply system using the IBBS20L to provide power backup. ● **477** is recommended for a power supply system using the ESM-48100B1/ESM-48150B1/ ESM-48100U2 to provide power backup. ● **461** is recommended for the APM30H (Ver.B)/ APM30H (Ver.C)/APM30H (Ver.D)/APM30H (Ver.E) cabinet using lithium batteries for power backup.

Parameter Name	Parameter ID	Setting Notes
Associated RXU Board CN	**BTSAPMUBP**.***ASSORXUCN***	This parameter specifies the cabinet number of the RRU connecting to the PMU when the PMU is installed on the RRU side.
Associated RXU Board SRN	**BTSAPMUBP**.***ASSORXUSRN***	This parameter specifies the subrack number of the RRU connecting to the PMU when the PMU is installed on the RRU side.
Associated RXU Board SN	**BTSAPMUBP**.***ASSORXUSN***	This parameter specifies the slot number of the RRU connecting to the PMU when the PMU is installed on the RRU side.
Manager Cabinet No.	**BTSAPMUBP**.***MCN***	The PMU is managed either by a BBU, CCU, or an RRU. Set this parameter to the cabinet number of the BBU, CCU, or RRU.
Manager Subrack No.	**BTSAPMUBP**.***MSRN***	The PMU is managed either by a BBU, CCU, or an RRU. Set this parameter to the subrack number of the BBU, CCU, or RRU.
Manager Port No.	**BTSAPMUBP**.***MPN***	• If the PMU is connected to the MON1 port on the UPEU, set this parameter to **1**. • If the PMU is connected to the MON0 port on the UPEU, set this parameter to **0**. • If the PMU is connected to an RRU, set this parameter to **0**. • If the PMU is connected to a CCU, set this parameter to **2**.
Address	**BTSAPMUBP**.***ADDR***	This parameter must be consistent with the value of the DIP switch on the PMU. The PMU supports DIP switch values of **3** (default value), **4**, and **9**[1]. To modify the communication address, you must modify both the parameter value and the value of the DIP switch.

NOTE

[1]: Set the PMU communication address as follows:

- Set the communication address of a single PMU to **3**.
- When PMUs are configured:
  - If two PMUs are configured on different RS485 buses, set the communication addresses to **3**.
  - If two PMUs are cascaded on one RS485 bus, set the communication address of the upper-level PMU to **3** and set the communication address of the lower-level PMU to **4**.
- There is no DIP switch for the PMU of the OPM50M. The BBU or RRU monitors the cascaded position of the OPM50M to automatically generate a communication address for the PMU.
  - If the OPM50M is monitored by a BBU3910A, there can only be a maximum of three-level OPM50M cascading on the RS485 bus. Set the communication address of the upper-level PMU to **3**, set that of the medium-level PMU to **4**, and set that of the lower-level PMU to **9**.
  - If the OPM50M is monitored by an RRU, cascading is not supported, and the communication address is **3** by default.
- The OPM30M does not support cascading connection, and the communication address is **3** by default.
- When PMUs are configured in a BDU:
  - If two PMUs are configured on different RS485 buses, set the communication addresses to **16**.
  - If two PMUs are cascaded on one RS485 bus, set the communication address of the upper-level PMU to **16** and set the communication address of the lower-level PMU to **17**.
  - When the PMU 12A connects to the MON0 or MON1 bus through an RS485 port, two PMU 12A modules must be cascaded before being connected to the bus to avoid PMU address conflict.
  - If the PMU is remotely monitored through the RRU power cable, set the **Manager Port No.** parameter to **0** and the communication address to **16**.
  - Set the communication address of the PMU 15A/PMU 15A (Ver.B) to **4**.

[2]: The end-of-discharge voltage of the lithium battery is higher than that of the lead-acid battery. If the **BTSAPMUBP**.***DCVLTHD*** parameter is set to the value recommended for the lead-acid battery in a lithium battery backup scenario, the base station cannot report ALM-25621 Power Supply DC Output Out of Range properly.

Scenario 2: PSU Configuration

Table 4-16 describes the parameter settings for PSU configuration.

Table 4-16 Parameter settings for PSU configuration

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	**BTSBRD**.***CN***	Set this parameter based on the actual situation.
Subrack No.	**BTSBRD**.***SRN***	Generally, the default value is used.

Parameter Name	Parameter ID	Setting Notes
Slot No.	**BTSBRD**.***SN***	• Configure the PSU in any of slots 1 to 3 in the APM30H (Ver.C). • Configure the PSU in any of slots 1 to 5 in the APM30H (Ver.D) or IBC10 AC cabinet. • Configure the PSU in any of slots 1 to 5 in the APM30H (Ver.E) AC cabinet. • Configure the PSU in slot 1 or 2 in the OMB (Ver.C). • Configure the PSU in any of slots 1 to 7 in the BTS3900AL (Ver.A) or TP48600A-H17B1. • Configure the PSU in any of slots 1 to 3 in the AC BTS3900 (Ver.C), BTS3900 (Ver.D), or PS4890. • The PSU cannot be configured when the power supply system type is SC48200 or SC4850. • Configure the PSU in any of slots 1 to 3 in an MTS or ICC power cabinet.
Board Type	**BTSBRD**.***BT***	Set this parameter to **PSU**.

Scenario 3: BDU Configuration

Table 4-17 describes the parameter settings for BDU configuration.

Table 4-17 Parameter settings for BDU configuration

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	**BTSBRD**.***CN***	Set this parameter based on the actual situation.
Subrack No.	**BTSBRD**.***SRN***	Generally, the default value is used.
Slot No.	**BTSBRD**.***SN***	Configure the BDU in slot 1 or 2 in the EPU02D or EPU02D-02.

Scenario 4: Storage Battery Configuration

NOTE

- If the EPS4815 PMU is configured, no storage battery can be configured.
- If the SC48200 PMU is configured, storage batteries are configured automatically by default.
- If other PMUs are configured, storage batteries are configured as required.
- The configured battery parameters should be consistent with the actual situation. Otherwise, storage batteries may be damaged.

Table 4-18 describes the parameter settings for storage battery configuration.

Table 4-18 Parameter settings for storage battery configuration

Parameter Name	Parameter ID	Setting Notes
Battery Configuration Enabled	**BTSAPMUBP.*BE***	**NO** is recommended.
Battery Type	**BTSAPMUBP.*BTYPE***	Only lead acid batteries and lithium batteries are available. ● Set this parameter to **NO_BAT** if no storage battery is installed. ● Set this parameter to **LI_BAT** if lithium batteries are installed. ● Set this parameter to **VRLA_INNER_BAT** if lead acid batteries are installed in the power supply cabinet or to **VRLA_OUTER_BAT** if they are installed in an independent battery cabinet.
Battery Number	**BTSAPMUBP.*BN***	This parameter is valid only when the **BTYPE** parameter is set to **LI_BAT**.
Battery Current Limiting Coefficient	**BTSAPMUBP.*BCLC***	This parameter specifies the battery charging current limiting coefficient. The maximum charging current can be obtained using the following formula: Maximum charging current = Battery charging current limiting coefficient x Battery capacity. If the charging current is greater than the maximum charging current by 5 A or above, ALM-25625 Battery Current Out of Range is reported. The value **15** is recommended.
Battery Capacity	**BTSAPMUBP.*BC***	Set this parameter based on the actual situation. Incorrect settings may affect the lifetime of the batteries. For the requirements on the capacity configuration, see the product manual delivered with the storage batteries.
Battery Number	**BTSAPMUBP.*BN***	This parameter is valid only when the **BTYPE** parameter is set to **LI_BAT**. The value **1** is recommended.

Parameter Name	Parameter ID	Setting Notes
Boost-Charging Voltage	**BTSAPMUBP**.***BCV***	This parameter specifies the boost-charging voltage. Typically, boost charging is used to quickly recover the capacity of the storage batteries. Therefore, the boost-charging voltage is relatively high. After the storage batteries are discharged, boost charging is conducted automatically. The value **565** is recommended.
Float-Charging Voltage	**BTSAPMUBP**.***FCV***	This parameter specifies the float charging voltage. A float charging is used to compensate for the power loss of the storage batteries due to self-discharge. The float charging voltage is generally lower than the boost charging voltage. **535** is recommended.
Shutdown Voltage	**BTSAPMUBP**.***SDV***	● If lead-acid batteries are used to provide power backup: – **460** is recommended for the SC48200 power system. – **430** is recommended for other power systems. ● If lithium batteries are used to provide power backup[2]: – **482** is recommended for a power supply system using the IBBS20L to provide power backup. – **462** is recommended for a power supply system using the ESM-48100B1/ESM-48150B1/ESM-48100U2 to provide power backup. – **430** is recommended for the APM30H (Ver.B)/APM30H (Ver.C)/APM30H (Ver.D)/APM30H (Ver.E) cabinet using lithium batteries for power backup.
Battery Temperature Compensated Configure Enabled	**BTSAPMUBP**.***CELLTEMPCOMPENABLED***	**NO** is recommended.
Upper Assembled Battery 1 Temp	**BTSAPMUBP**.***CELLTEMP1THRESHOLDH***	Battery group temperature sensor 1 is installed inside the cabinet. **800** is recommended.
Lower Assembled Battery1 TEMP Measure	**BTSAPMUBP**.***CELLTEMP1THRESHOLDL***	**-200** is recommended.

Parameter Name	Parameter ID	Setting Notes
Temperature Alarm Upper Threshold	**BTSAPMUBP**.*TUTHD*	This parameter specifies the upper temperature alarm threshold. When the battery temperature exceeds the threshold, ALM-25650 Environmental Temperature Abnormal is reported. The value **50** is recommended.
Temperature Alarm Lower Threshold	**BTSAPMUBP**.*TLTHD*	This parameter specifies the lower temperature alarm threshold. When the battery temperature is lower than the threshold, ALM-25650 Environmental Temperature Abnormal is reported. **0** is recommended for the EPS4815 power system, and **–19** is recommended for other power systems.
Temperature Compensation Coefficient	**BTSAPMUBP**.*TCC*	**80** is recommended.
Temperature Basis for Compensation	**BTSAPMUBP**.*BASETEMPERATURE*	**Battery1Temp** is recommended for the EPS4815 power system.
Battery Discharge Depth	**BTSAPMUBP**.*BATTERYDISCHARGEDEPTH*	The value **50** is recommended.
Boost-Charging Duration	**BTSAPMUBP**.*BCD*	For the specific configuration, see the product manual delivered with the storage batteries.

NOTE

[2]: The end-of-discharge voltage of the lithium battery is higher than that of the lead-acid battery. If the **BTSAPMUBP**.*DCVLTHD* parameter is set to the value recommended for the lead-acid battery in a lithium battery backup scenario, the base station cannot report ALM-25621 Power Supply DC Output Out of Range properly.

Scenario 5: Diesel Generator Configuration

Table 4-19 describes parameter settings for diesel generator configuration.

Table 4-19 Parameter settings for diesel generator configuration

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	**BTSAPMUBP**.*CN*	Set this parameter based on the actual situation.
Subrack No.	**BTSAPMUBP**.*SRN*	Generally, the default value is used.

Parameter Name	Parameter ID	Setting Notes
Slot No.	**BTSAPMUBP**.***SN***	Generally, it is set to **0**.
Rated Power	**BTSAPMUBP**.***POWER***	Set this parameter based on the rated output power printed on the name plate of the diesel generator. **125** is recommended.
Diesel Engine Configure Enabled	**BTSAPMUBP**.***SETDIESELENGINEENABLED***	When this parameter is set to **YES**, the **BTSAPMUBP**.***POWER*** parameter can be configured.

4.5.1.2 Using MML Commands

Before using MML commands, refer to **4.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

4.5.1.2.1 Activation Command Examples

NOTE

The following scenarios can exist simultaneously. If they do, MOs for all these scenarios must be configured.

Scenario 1: PMU Configuration

Step 1 Run the **ADD BTSBRD** command to add a PMU with the cabinet number, subrack number, and slot number set to correct values.

Step 2 Run the **SET BTSAPMUBP** command. In this step, set the parameters in **Table 4-15**.

----End

NOTE

- The **BTSBRD**.***MCN***, **BTSBRD**.***MSRN***, **BTSBRD**.***MPN***, and **BTSBRD**.***ADDR*** parameters must be correctly specified to ensure proper communication between the PMU and BBU.
- **PTYPE** must be correctly specified. Then, set the **BTYPE**, **BCLC**, and **BC** parameters based on the actual situation. Incorrectly setting the **BC** parameter reduces the lifetime of the storage batteries.

Scenario 2: PSU Configuration

Step 1 Run the **ADD BTSBRD** command to add a PSU with the cabinet number, subrack number, and slot number set to correct values. In this step, set the parameters in **Table 4-16**.

----End

Scenario 4: Storage Battery Configuration

Step 1 Run the **SET BTSAPMUBP** command. In this step, set the parameters in **Table 4-18**.

----End

Scenario 5: Diesel Generator Configuration

Step 1 Run the **SET BTSAPMUBP** command. In this step, set the parameters in **Table 4-19**. The methods for configuring one or two diesel generators are the same.

----End

4.5.1.2.2 Deactivation Command Examples

N/A

4.5.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.5.2 Activation Verification

Method 1: Run the **DSP BTSBRD** command to query the basic information of each board in the power system and check whether the board status is normal.

Method 2: Check whether power devices such as the PMU, PSU, and PDB are normal on the MAE-Access or Web LMT device panel.

NOTE

The power devices in the PMU subrack, such as the PSU and PDB, are managed by the PMU. The alarm status of these devices is displayed under the PMU.

4.6 Operation and Maintenance (Multimode Base Station)

A multimode base station can be deployed as a separate-MPT or co-MPT base station.

- The deployment of this feature on the co-MPT multimode base station is the same as that on the eGBTS, NodeB, eNodeB, or gNodeB. For details, see "Operation and Maintenance (eGBTS/NodeB/eNodeB/gNodeB)" in section 4.4.
- For the separate-MPT multimode base station, this feature can be configured in either one-sided mode or dual-sided mode. One-sided configuration is recommended.

NOTE

One-sided configuration has been available since SRAN6.0. Versions earlier than SRAN6.0 support only dual-sided configuration. RRU remote monitoring supports only one-sided configuration.

- When one-sided configuration is used, configure the PMU for only the RAT managing the PMU. The configuration method is the same as that for a single-mode base station.
- Dual-sided configuration does not require a specific configuration sequence. You only need to follow the instructions in **4.5 Operation and Maintenance (GBTS)** or **4.4 Operation and Maintenance (eGBTS/NodeB/eNodeB/gNodeB)** to perform the configurations for both RATs and ensure the consistency of parameters between RATs.
- **Table 4-20** to **Table 4-24** list the parameters of which the settings must be consistent between RATs in a GBTS. **Table 4-25** to **Table 4-29** list the parameters of which the settings must be consistent between RATs in an eGBTS/NodeB/eNodeB/gNodeB. The configuration object is the PMU.

Table 4-20 Common parameters of the multimode base stations–PMU, GBTS (1)

Parameter Name	Parameter ID	MML Command	Setting Notes
Power Type	**BTSAPMUBP.*PTYPE***	**SET BTSAPMUBP**	For the mapping between cabinet types and power supply system types, see **Table 4-5**, **Table 4-6**, and **Table 4-7**.
AC Voltage Alarm Lower Threshold	**BTSAPMUBP.*ACVLTHD***	**SET BTSAPMUBP**	**180** is recommended.
AC Voltage Alarm Upper Threshold	**BTSAPMUBP.*ACVUTHD***	**SET BTSAPMUBP**	**280** is recommended.
DC Voltage Alarm Upper Threshold	**BTSAPMUBP.*DCVUTHD***	**SET BTSAPMUBP**	**585** is recommended for the SC48200 PMU. **580** is recommended for other power supply systems. ● This parameter can be set to a value ranging from **530** to **600** when **PMU.*PTYPE*** is set to **SMU**. ● This parameter can be set to a value ranging from **580** to **600** when **PMU.*PTYPE*** is set to a value other than **SMU**.

Parameter Name	Parameter ID	MML Command	Setting Notes
DC Voltage Alarm Lower Threshold	**BTSAPMUBP**.***DCVLTHD***	**SET BTSAPMUBP**	If lead-acid batteries are used to provide power backup: ● **470** is recommended for the SC48200 PMU. ● **450** is recommended for other power supply systems. If lithium batteries are used to provide power backup[2]: ● **500** is recommended for a power supply system using the IBBS20L to provide power backup. ● **477** is recommended for a power supply system using the ESM-48100B1/ESM-48150B1/ESM-48100U2 to provide power backup. ● **461** is recommended for the APM30H (Ver.B)/APM30H (Ver.C)/APM30H (Ver.D)/APM30H (Ver.E) cabinet using lithium batteries for power backup.
Load Shutdown Flag	**BTSAPMUBP**.***LSDF***	**SET BTSAPMUBP**	**ENABLE** is recommended for the SC48200 power supply system. **DISABLE** is recommended for other power supply systems.

Parameter Name	Parameter ID	MML Command	Setting Notes
Load Shutdown Voltage	**BTSAPMUBP**.***LSDV***	**SET BTSAPMUBP**	If lead-acid batteries are used to provide power backup: ● **465** is recommended for the SC48200 power supply system. ● **440** is recommended for other power supply systems. If lithium batteries are used to provide power backup[1]: ● **495** is recommended for a power supply system using the IBBS20L to provide power backup. ● **476** is recommended for a power supply system using the ESM-48100B1/ESM-48150B1/ESM-48100U2 to provide power backup. ● **460** is recommended for the APM30H (Ver.B)/APM30H (Ver.C)/APM30H (Ver.D)/APM30H (Ver.E) cabinet using lithium batteries for power backup.
Load Shutdown Time Flag	**BTSAPMUBP**.***LSDTF***	**SET BTSAPMUBP**	The default value is recommended.
Load Shutdown Time	**BTSAPMUBP**.***LSDT***	**SET BTSAPMUBP**	The default value is recommended.

NOTE

[2]: The end-of-discharge voltage of the lithium battery is higher than that of the lead-acid battery. If the **BTSAPMUBP**.***DCVLTHD*** parameter is set to the value recommended for the lead-acid battery in a lithium battery backup scenario, the base station cannot report ALM-25621 Power Supply DC Output Out of Range properly.

[1]: The end-of-discharge voltage of the lithium battery is higher than that of the lead-acid battery. If the ***LSDV*** parameter is set to the recommended value for the lead-acid battery in a lithium battery backup scenario, discharge undervoltage protection will be performed on the lithium battery first. Then, load power-off will not take effect when it is enabled.

Table 4-21 Common parameters of the multimode base stations–PMU, GBTS (2)

Parameter Name	Parameter ID	MML Command	Setting Notes
Load Shutdown Close BBU board	**BTSOTHPARA**.***SDBBLSD***	**SET BTSOTHPARA**	Set this parameter based on site requirements.
Special Analog Alarm Flag	**BTSAPMUBP**.***SAAF***	**SET BTSAPMUBP**	● You are not advised to configure any sensor on the PMU. Therefore, all sensors are generally disabled. That is, the parameter is set to **11111**. ● Battery temperature sensor 1 is enabled when the power supply system is PS4890 or SC48200 or when storage batteries are configured. It is disabled by default when any other power supply system is configured, such as the ETP.
Special Boolean Alarm Flag	**BTSAPMUBP**.***SBAF***	**SET BTSAPMUBP**	It is set to the disabled state for the water sensor and smoke sensor.
Battery Type	**BTSAPMUBP**.***BTYPE***	**SET BTSAPMUBP**	Set this parameter to **OUTER** for the SC48200 PMU. Set this parameter based on the actual situation for other PMUs.
Battery Type	**BTSAPMUBP**.***BTYPE***	**SET BTSAPMUBP**	● **VRLA_BAT** is recommended for UMTS/LTE. ● **VRLA_OUTER_BAT** is recommended for GSM.
Battery Capacity	**BTSAPMUBP**.***BC***	**SET BTSAPMUBP**	For the requirements on the capacity configuration, see the product manual delivered with the storage batteries.
Float-Charging Voltage	**BTSAPMUBP**.***FCV***	**SET BTSAPMUBP**	**535** is recommended.
Boost-Charging Duration	**BTSAPMUBP**.***BCD***	**SET BTSAPMUBP**	For the specific configuration, see the product manual delivered with the storage batteries.
Boost-Charging Voltage	**BTSAPMUBP**.***BCV***	**SET BTSAPMUBP**	**565** is recommended.

Table 4-22 Common parameters of the multimode base stations–PMU, GBTS (3)

Parameter Name	Parameter ID	MML Command	Setting Notes
Battery Current Limiting Coefficient	**BTSAPMUBP**.***BCLC***	**SET BTSAPMUBP**	**15** is recommended.
Temperature Compensation Coefficient	**BTSAPMUBP**.***TCC***	**SET BTSAPMUBP**	**80** is recommended.
Temperature Alarm Lower Threshold	**BTSAPMUBP**.***TLTHD***	**SET BTSAPMUBP**	**–19** is recommended.
Temperature Alarm Upper Threshold	**BTSAPMUBP**.***TUTHD***	**SET BTSAPMUBP**	**50** is recommended.
High Temperature Shutdown Flag	**BTSAPMUBP**.***HTSDF***	**SET BTSAPMUBP**	**ENABLE** is recommended.
Shutdown Temperature	**BTSAPMUBP**.***SDT***	**SET BTSAPMUBP**	**60** is recommended for the SC48200 PMU. **53** is recommended for other PMUs.
Low Voltage Shutdown Flag	**BTSAPMUBP**.***LVSDF***	**SET BTSAPMUBP**	**ENABLE** is recommended.

Parameter Name	Parameter ID	MML Command	Setting Notes
Shutdown Voltage	**BTSAPMUBP**.*SDV*	**SET BTSAPMUBP**	If lead-acid batteries are used to provide power backup: ● **460** is recommended for the SC48200 power supply system. ● **430** is recommended for other power supply systems. If lithium batteries are used to provide power backup[1]: ● **482** is recommended for a power supply system using the IBBS20L to provide power backup. ● **462** is recommended for a power supply system using the ESM-48100B1/ESM-48150B1/ESM-48100U2 to provide power backup. ● **430** is recommended for the APM30H (Ver.B)/APM30H (Ver.C)/APM30H (Ver.D)/APM30H (Ver.E) cabinet using lithium batteries for power backup.

NOTE

[1]: The end-of-discharge voltage of the lithium battery is higher than that of the lead-acid battery. If the default value for the lead-acid battery is used, discharge undervoltage protection will be performed on the lithium battery first. Then, power-off under low voltage will not take effect when it is enabled.

Table 4-23 Common parameters of the multimode base stations–PMU, GBTS (4)

Parameter Name	Parameter ID	MML Command	Setting Notes
0.05C10 Discharge Time	**BTSAPMUBP**.*DSCHGT0*	**SET BTSAPMUBP**	**1200** is recommended.
0.1C10 Discharge Time	**BTSAPMUBP**.*DSCHGT1*	**SET BTSAPMUBP**	**600** is recommended.
0.2C10 Discharge Time	**BTSAPMUBP**.*DSCHGT2*	**SET BTSAPMUBP**	**300** is recommended.

Parameter Name	Parameter ID	MML Command	Setting Notes
0.3C10 Discharge Time	**BTSAPMUBP**.***DSCHGT3***	**SET BTSAPMUBP**	**150** is recommended.
0.4C10 Discharge Time	**BTSAPMUBP**.***DSCHGT4***	**SET BTSAPMUBP**	**100** is recommended.
0.5C10 Discharge Time	**BTSAPMUBP**.***DSCHGT5***	**SET BTSAPMUBP**	**70** is recommended.
0.6C10 Discharge Time	**BTSAPMUBP**.***DSCHGT6***	**SET BTSAPMUBP**	**50** is recommended.
0.7C10 Discharge Time	**BTSAPMUBP**.***DSCHGT7***	**SET BTSAPMUBP**	**40** is recommended.
0.8C10 Discharge Time	**BTSAPMUBP**.***DSCHGT8***	**SET BTSAPMUBP**	**30** is recommended.
0.9C10 Discharge Time	**BTSAPMUBP**.***DSCHGT9***	**SET BTSAPMUBP**	**25** is recommended.

Table 4-24 Common parameters of the multimode base stations–PMU, GBTS (5)

Parameter Name	Parameter ID	MML Command	Setting Notes
Battery Charge Efficiency	**BTSAPMUBP**.***EFF***	**SET BTSAPMUBP**	**80** is recommended.
Discharge Test End Voltage	**BTSAPMUBP**.***ENDV***	**SET BTSAPMUBP**	**190** is recommended.
Discharge Test Time Limit	**BTSAPMUBP**.***DSTML***	**SET BTSAPMUBP**	**10** is recommended.
Simple Discharge Test End Voltage	**BTSAPMUBP**.***SDSEV***	**SET BTSAPMUBP**	**450** is recommended.

Parameter Name	Parameter ID	MML Command	Setting Notes
Simple Discharge Test Time Limit	**BTSAPMUBP.*SDSTML***	**SET BTSAPMUBP**	**60** is recommended.
Timing Discharge Test Time	**BTSAPMUBP.*TDSTM***	**SET BTSAPMUBP**	**120** is recommended.
Delayed Discharge Test Time	**BTSAPMUBP.*DDSTM***	**SET BTSAPMUBP**	**14** is recommended.
Rated Power	**BTSAPMUBP.*POWER***	**SET BTSAPMUBP**	**125** is recommended.

Table 4-25 Common parameters of the multimode base stations–PMU, eGBTS/NodeB/eNodeB/gNodeB (6)

Parameter Name	Parameter ID	MML Command	Setting Notes
Power System Type	**PMU.*PTYPE (LTE eNodeB, 5G gNodeB)***	**ADD PMU**	For the mapping between cabinet types and power supply system types, see **Table 4-5**, **Table 4-6**, and **Table 4-7**.
AC Voltage Alarm Lower Threshold	**PMU.*ACVLTHD (LTE eNodeB, 5G gNodeB)***	**ADD PMU**	**180** is recommended.
AC Voltage Alarm Upper Threshold	**PMU.*ACVUTHD (LTE eNodeB, 5G gNodeB)***	**ADD PMU**	**280** is recommended.
DC Voltage Alarm Upper Threshold	**PMU.*DCVUTHD (LTE eNodeB, 5G gNodeB)***	**ADD PMU**	**585** is recommended for the SC48200 PMU. **580** is recommended for other power supply systems. ● This parameter can be set to a value ranging from **530** to **600** when **PMU.*PTYPE*** is set to **SMU**. ● This parameter can be set to a value ranging from **580** to **600** when **PMU.*PTYPE*** is set to a value other than **SMU**.

Parameter Name	Parameter ID	MML Command	Setting Notes
DC Voltage Alarm Lower Threshold	**PMU**.***DCVLTHD (LTE eNodeB, 5G gNodeB)***	**ADD PMU**	If lead-acid batteries are used to provide power backup: ● **470** is recommended for the SC48200 PMU. ● **450** is recommended for other PMUs. If lithium batteries are used to provide power backup[1]: ● **500** is recommended for a power supply system using the IBBS20L to provide power backup. ● **477** is recommended for a power supply system using the ESM-48100B1/ESM-48150B1/ESM-48100U2 to provide power backup. ● **461** is recommended for the APM30H (Ver.B)/APM30H (Ver.C)/APM30H (Ver.D)/APM30H (Ver.E) cabinet using lithium batteries for power backup. ● When **PMU**.***PTYPE*** is set to **SMU**, **PMU**.***DCVLTHD*** must be set to a value greater than or equal to **360**.
Load Shutdown Flag	**PMU**.***LSDF (LTE eNodeB, 5G gNodeB)***	**ADD PMU**	● **ENABLE** is recommended for the SC48200 PMU. ● **DISABLE** is recommended for other PMUs.

Parameter Name	Parameter ID	MML Command	Setting Notes
Load Shutdown Voltage	**PMU**.***LSDV (LTE eNodeB, 5G gNodeB)***	**ADD PMU**	If lead-acid batteries are used to provide power backup: ● **465** is recommended for the SC48200 power supply system. ● **440** is recommended for other power supply systems. If lithium batteries are used to provide power backup[2]: ● **495** is recommended for a power supply system using the IBBS20L to provide power backup. ● **476** is recommended for a power supply system using the ESM-48100B1/ESM-48150B1/ESM-48100U2 to provide power backup. ● **460** is recommended for the APM30H (Ver.B)/APM30H (Ver.C)/APM30H (Ver.D)/APM30H (Ver.E) cabinet using lithium batteries for power backup. ● When **PMU**.***PTYPE*** is set to **SMU** and **PMU**.***LSDF*** is set to **ENABLE**, **PMU**.***LSDV*** must be set to a value greater than or equal to **360**.
Load Shutdown Time Flag	**PMU**.***LSDTF (LTE eNodeB, 5G gNodeB)***	**ADD PMU**	The default value is recommended.
Load Shutdown Time	**PMU**.***LSDT (LTE eNodeB, 5G gNodeB)***	**ADD PMU**	The default value is recommended.

NOTE

[1]: The end-of-discharge voltage of the lithium battery is higher than that of the lead-acid battery. If the ***DCVLTHD*** parameter is set to the value recommended for the lead-acid battery in a lithium battery backup scenario, the base station cannot report ALM-25621 Power Supply DC Output Out of Range properly.

[2]: The end-of-discharge voltage of the lithium battery is higher than that of the lead-acid battery. If the ***LSDV*** parameter is set to the recommended value for the lead-acid battery in a lithium battery backup scenario, discharge undervoltage protection will be performed on the lithium battery first. Then, load power-off will not take effect when it is enabled.

Table 4-26 Common parameters of the multimode base stations–PMU, eGBTS/NodeB/eNodeB/gNodeB (7)

Parameter Name	Parameter ID	MML Command	Setting Notes
Shut Down BBU Boards in Load Power Off	**EQUIPMENT**.***SDBBLSD (LTE eNodeB, 5G gNodeB)***	**SET EQUIPMENT**	Set this parameter based on site requirements.
Special Analog Alarm Flag	**PMU**.***SAAF (LTE eNodeB, 5G gNodeB)***	**ADD PMU**	Set this parameter based on the actual situation. You are not advised to configure any sensor on the PMU. Therefore, the ***SAAF*** parameter is generally disabled.
Special Boolean Alarm Flag	**PMU**.***SBAF (LTE eNodeB, 5G gNodeB)***	**ADD PMU**	Set this parameter based on the actual situation.
Installation Type	**BATTERY**.***INSTALLTYPE (LTE eNodeB, 5G gNodeB)***	**ADD BATTERY**	Set this parameter to **OUTER** for the SC48200 PMU. Set this parameter based on the actual situation for other PMUs.
Battery Type	**BATTERY**.***BTYPE (LTE eNodeB, 5G gNodeB)***	**ADD BATTERY**	● **VRLA_BAT** is recommended for UMTS/LTE/NR. ● **VRLA_OUTER_BAT** is recommended for GSM.
Battery 1 Capacity Battery 2 Capacity	**BATTERY**.***BC1 (LTE eNodeB, 5G gNodeB)*** ***BC2 (LTE eNodeB, 5G gNodeB)***	**ADD BATTERY**	For the requirements on the capacity configuration, see the product manual delivered with the storage batteries.
Float-Charging Voltage	**BATTERY**.***FCV (LTE eNodeB, 5G gNodeB)***	**ADD BATTERY** **MOD BATTERY**	**535** is recommended.
Boost-Charging Duration	**BATTERY**.***BCD (LTE eNodeB, 5G gNodeB)***	**ADD BATTERY**	For the specific configuration, see the product manual delivered with the storage batteries.
Boost-Charging Voltage	**BATTERY**.***BCV (LTE eNodeB, 5G gNodeB)***	**ADD BATTERY**	**565** is recommended.

Table 4-27 Common parameters of the multimode base stations–PMU, eGBTS/NodeB/eNodeB/gNodeB (8)

Parameter Name	Parameter ID	MML Command	Setting Notes
Battery Current Limiting Coefficient	**BATTERY**.***BCLC (LTE eNodeB, 5G gNodeB)***	**ADD BATTERY**	**15** is recommended.
Temperature Compensation Coefficient	**BATTERY**.***TCC (LTE eNodeB, 5G gNodeB)***	**ADD BATTERY**	**80** is recommended.
Temperature Alarm Lower Threshold	**BATTERY**.***TLTHD (LTE eNodeB, 5G gNodeB)***	**ADD BATTERY**	**–19** is recommended.
Temperature Alarm Upper Threshold	**BATTERY**.***TUTHD (LTE eNodeB, 5G gNodeB)***	**ADD BATTERY**	**50** is recommended.
High Temperature Shutdown Flag	**BATTERY**.***HTSDF (LTE eNodeB, 5G gNodeB)***	**ADD BATTERY**	**ENABLE** is recommended.
Shutdown Temperature	**BATTERY**.***SDT (LTE eNodeB, 5G gNodeB)***	**ADD BATTERY**	**60** is recommended for the SC48200 PMU. **53** is recommended for other PMUs.
Low Voltage Shutdown Flag	**BATTERY**.***LVSDF (LTE eNodeB, 5G gNodeB)***	**ADD BATTERY**	**ENABLE** is recommended.

Parameter Name	Parameter ID	MML Command	Setting Notes
Shutdown Voltage	**BATTERY**.***SDV (LTE eNodeB, 5G gNodeB)***	**ADD BATTERY**	If lead-acid batteries are used to provide power backup: ● **460** is recommended for the SC48200 PMU. ● **430** is recommended for other PMUs. If lithium batteries are used to provide power backup[1]: ● **482** is recommended for a power supply system using the IBBS20L to provide power backup. ● **462** is recommended for a power supply system using the ESM-48100B1/ESM-48150B1/ESM-48100U2 to provide power backup. ● **430** is recommended for the APM30H (Ver.B)/APM30H (Ver.C)/APM30H (Ver.D)/APM30H (Ver.E) cabinet using lithium batteries for power backup. ● When an MTS or ICC power cabinet uses lithium batteries for power backup, this parameter can be set to a value ranging from **360** to **556**. The value **432** is recommended.

NOTE

[1]: The end-of-discharge voltage of the lithium battery is higher than that of the lead-acid battery. If the default value for the lead-acid battery is used, discharge undervoltage protection will be performed on the lithium battery first. Then, power-off under low voltage will not take effect when it is enabled.

Table 4-28 Common parameters of the multimode base stations–PMU, eGBTS/NodeB/eNodeB/gNodeB (9)

Parameter Name	Parameter ID	MML Command	Setting Notes
0.05C10 Discharge Time	**BATCTPA**.***DSCHGT0 (LTE eNodeB, 5G gNodeB)***	**MOD BATCTPA**	**1200** is recommended.

Parameter Name	Parameter ID	MML Command	Setting Notes
0.1C10 Discharge Time	**BATCTPA**.***DSCHGT1 (LTE eNodeB, 5G gNodeB)***	**MOD BATCTPA**	**600** is recommended.
0.2C10 Discharge Time	**BATCTPA**.***DSCHGT2 (LTE eNodeB, 5G gNodeB)***	**MOD BATCTPA**	**300** is recommended.
0.3C10 Discharge Time	**BATCTPA**.***DSCHGT3 (LTE eNodeB, 5G gNodeB)***	**MOD BATCTPA**	**150** is recommended.
0.4C10 Discharge Time	**BATCTPA**.***DSCHGT4 (LTE eNodeB, 5G gNodeB)***	**MOD BATCTPA**	**100** is recommended.
0.5C10 Discharge Time	**BATCTPA**.***DSCHGT5 (LTE eNodeB, 5G gNodeB)***	**MOD BATCTPA**	**70** is recommended.
0.6C10 Discharge Time	**BATCTPA**.***DSCHGT6 (LTE eNodeB, 5G gNodeB)***	**MOD BATCTPA**	**50** is recommended.
0.7C10 Discharge Time	**BATCTPA**.***DSCHGT7 (LTE eNodeB, 5G gNodeB)***	**MOD BATCTPA**	**40** is recommended.
0.8C10 Discharge Time	**BATCTPA**.***DSCHGT8 (LTE eNodeB, 5G gNodeB)***	**MOD BATCTPA**	**30** is recommended.
0.9C10 Discharge Time	**BATCTPA**.***DSCHGT9 (LTE eNodeB, 5G gNodeB)***	**MOD BATCTPA**	**25** is recommended.

Table 4-29 Common parameters of the multimode base stations–PMU, eGBTS/NodeB/eNodeB/gNodeB (10)

Parameter Name	Parameter ID	MML Command	Setting Notes
Battery Charge Efficiency	**BATCTPA**.***EFF (LTE eNodeB, 5G gNodeB)***	**MOD BATCTPA**	**80** is recommended.
Discharge Test End Voltage	**BATCTPA**.***ENDV (LTE eNodeB, 5G gNodeB)***	**MOD BATCTPA**	**190** is recommended.

Parameter Name	Parameter ID	MML Command	Setting Notes
Discharge Test Time Limit	**BATCTPA**.***DSTML (LTE eNodeB, 5G gNodeB)***	**MOD BATCTPA**	**10** is recommended.
Simple Discharge Test End Voltage	**BATCTPA**.***SDSEV (LTE eNodeB, 5G gNodeB)***	**MOD BATCTPA**	**450** is recommended. When **PMU**.***PTYPE*** is set to **SMU**, the **BATCTPA**.***SDSEV*** parameter must be set to a value greater than or equal to **442** and less than or equal to **530**.
Simple Discharge Test Time Limit	**BATCTPA**.***SDSTML (LTE eNodeB, 5G gNodeB)***	**MOD BATCTPA**	**60** is recommended.
Timing Discharge Test Time	**BATCTPA**.***TDSTM (LTE eNodeB, 5G gNodeB)***	**MOD BATCTPA**	**120** is recommended.
Delayed Discharge Test Time	**BATCTPA**.***DDSTM (LTE eNodeB, 5G gNodeB)***	**MOD BATCTPA**	**14** is recommended.
Rated Power	**DIESELGEN**.***POWER (LTE eNodeB, 5G gNodeB)***	**ADD DIESELGEN** **MOD DIESELGEN**	**125** is recommended.

4.7 Network Monitoring

None

5 Intelligent Battery Management

5.1 Principles

Intelligent battery management includes automatic switching between different charge-and-discharge modes, self-protection under high temperature, and battery runtime display. These functions enable the base station to prolong the battery lifetime by battery self-protection, and therefore increase profits of operators by reducing operating expense.

Intelligent battery management applies only to the following cabinets or power modules:

- APM30H (Ver.C)
- APM30H (Ver.D)
- APM30H (Ver.E)
- BTS3900AL (Ver.A)
- TP48600A-H17B1
- IBC10
- IBC10 (Ver.B)
- OMB (Ver.C)
- OPM200/OPM200 (Ver.B)
- OPM50M
- OPM50M (Ver.B)
- OPM30A
- APM5930(AC) (Ver.A)
- APM5950H (Ver.A)
- APM5950H-L (Ver.A)

This function applies to GSM, UMTS, NR, and LTE. The following sections describe the functions of intelligent battery management.

5.1.1 Automatic Switching Between Different Charge-and-Discharge Modes

Without automatic switching between different charge-and-discharge modes, fully charged batteries enter and stay in the float charging state to maintain electricity. The batteries receive much more electricity than they send out. This may corrode anodes, dry electrolytes, and shorten the battery lifetime.

This function enables a base station to switch the charge-and-discharge mode for batteries depending on the quality of the power grid. If the power grid quality is favorable, batteries enter and stay in the hibernation state, in which the batteries do not charge or discharge. This helps prolong the battery lifetime.

With this function, the base station calculates the number of times the mains supply fails and the duration of each instance over the past 15 days. Then, it determines the grid type and activates the corresponding charge-and-discharge mode for batteries. **Table 5-1** provides details about charge-and-discharge modes.

Table 5-1 Charge-and-discharge modes

Number of Accumulated Grid Abnormal Hours (in 15 Days)	Grid Type	Charge-and-Discharge Mode	Current Limit Value (C)	Hibernation Voltage (V)	Hibernation Duration (Days)	Estimated Improvement Rate of Battery Lifespan
Less than or equal to 5	1	Mode A	0.10	52	13	100%
More than 5 and less than or equal to 30	2	Mode B	0.15	52	6	50%
More than 30 and less than 120	3	Mode C	0.15	None	None	0%
More than 120	4	Mode C		None	None	

To enable automatic switching between different charge-and-discharge modes, obtain and activate the license for the feature and configure the ***BATIMS (LTE eNodeB, 5G gNodeB)*** parameter.

NOTE

- Boost charging: A state in which batteries can be fully charged at a higher voltage than normal within a short period. This state ensures that all the batteries are equally charged.
- Float charging: A state in which batteries are consistently charged at a constant voltage over a long period. This state compensates for self-discharge to ensure that the batteries are always fully charged.
- The current limit value limits the maximum charging current for the battery as follows: Maximum charging current = Current limit value x Battery capacity.

5.1.2 Self-Protection Under High Temperature

Under high temperature, the base station achieves automatic battery protection through methods such as switching the battery to the float charging state, lowering the charging voltage, or disconnecting the battery.

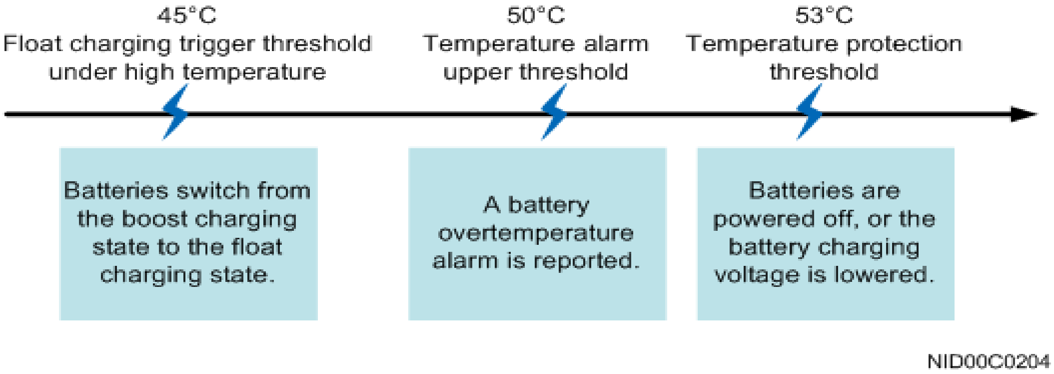
If this function is not enabled under high temperature, and the battery is charged or discharged in the same way as under normal temperature, the battery life cycle will be shortened or the battery will be damaged.

NOTICE

Under high temperature, this function may negatively affect the battery backup function. If batteries enter the float charging state or a base station lowers the battery charging voltage under high temperature, the batteries may not be fully charged and therefore the time the base station continues providing services using backup power is shortened. If the batteries are disconnected under high temperature, the base station has no backup power. Then, services on the base station will be interrupted if the base station uses the AC power input and the mains supply fails.

This function provides different solutions to prevent battery damage under different high temperatures, as shown in **Figure 5-1**. The figure assumes that the default configurations of the base station are in effect.

Figure 5-1 Solutions under different high temperatures



- If the battery temperature exceeds the threshold for entering the float charging state and remains above the threshold for 5 minutes, batteries enter the float charging state and no alarms are reported. This threshold equals to the value of the ***TUTHD (LTE eNodeB, 5G gNodeB)*** parameter minus 5.
- If the battery temperature remains higher than the value of ***TUTHD (LTE eNodeB, 5G gNodeB)*** for 5 minutes, a battery overtemperature alarm is reported.
- If the battery temperature remains higher than the value of ***SDT (LTE eNodeB, 5G gNodeB)*** for 5 minutes, the battery charging voltage is lowered or batteries are powered off based on the value of the ***HTSDF (LTE eNodeB, 5G gNodeB)*** parameter. Then, the batteries enter the high-temperature protection state.

- The ***TUTHD (LTE eNodeB, 5G gNodeB)***, ***SDT (LTE eNodeB, 5G gNodeB)***, and ***HTSDF (LTE eNodeB, 5G gNodeB)*** parameters apply to GBTSs, eGBTSs, NodeBs, eNodeBs, gNodeBs, and co-MPT base stations.

This function is enabled by default and not under license control. It cannot be disabled.

5.1.3 Battery Runtime Display

With the battery runtime display function, a base station calculates the battery runtime when the mains supply fails. Users can query the battery runtime anytime and prepare a diesel generator in advance (for example) to prevent service interruption and improve system reliability.

With this function, a base station uses the following formula to calculate the battery runtime when the mains supply is cut off and users can query the battery runtime by running the following MML commands:

- GBTS: **DSP BTSBATCAP**
- eGBTS/NodeB/eNodeB/gNodeB: **DSP BATTERY**

Battery runtime = (Remaining power capacity % x Total power capacity x Discharge efficiency)/(Mean discharge current x Aging coefficient)

NOTE

- The discharge efficiency is negatively correlated to the mean discharge current, and the aging coefficient is positively correlated to the time batteries have been in use.
- The calculated battery runtime may be inaccurate due to inaccurate calculation of the remaining power capacity and discharge current. The maximum error ranges from 10% to 20% when batteries are fully charged.

Querying the battery runtime by running an MML command is not under license control. It always works.

5.2 Network Analysis

5.2.1 Benefits

None

5.2.2 Impacts

None

5.3 Requirements

5.3.1 Licenses

To activate Intelligent Battery Management, purchase and activate the license for this feature. For a co-MPT base station, you only need to load and activate the license of one RAT.

Table 5-2 lists the license information about this feature on the eGBTS/NodeB/eNodeB. There are no license requirements on the gNodeB side.

Table 5-2 License information about this feature

RAT	Feature ID	Feature Name	Model	License Description	NE	Sales Unit
GSM	GBFD-510710	Intelligent Battery Management	LGMIIBM	Intelligent Battery Management (per Site)	eGBTS	per BTS
GSM	GBFD-510710	Intelligent Battery Management	LGMIIBM	Intelligent Battery Management (per Site)	GBSC	per BTS
UMTS	WRFD-140220	Intelligent Battery Management	LQW9IBATM01	Intelligent Battery Management (per NodeB)	NodeB	Per NodeB
LTE FDD	LOFD-001071	Intelligent Battery Management	LT1S000IBM00	Intelligent Battery Management(FDD)	eNodeB	Per Cell
LTE TDD	TDLOFD-001071	Intelligent Battery Management	LT1ST00IBM00	Intelligent Battery Management(TDD)	eNodeB	Per Cell
LTE NB-IoT	MLOFD-001071	Intelligent Battery Management	ML1S000IBM00	Intelligent Battery Management(NB-IoT)	eNodeB	Per Cell

5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.2.1 Prerequisite Functions

None

5.3.2.2 Mutually Exclusive Functions

None

5.3.3 Hardware

Base Station Models

- 3900 and 5900 series base stations (macro base stations)
- DBS3900 LampSite and DBS5900 LampSite

Power Supply Systems

For details, see **Power Supply** and **Mapping Between Physical Cabinets and PTYPE Values**.

5.3.4 Others

- Base station software version

  The base station software of GBSS14.0/RAN14.0/eRAN3.0/SRAN8.0/5G RAN1.0 or later is required.
- Power supply system type

  4.3.3 Hardware describes whether power supply systems support this function.
- PMU software version
  - If the power supply system type is ETP, the PMU software version must be 117 or later.
  - If the power supply system type is EPS4815, the PMU software version must be 128 or later.

5.4 Operation and Maintenance (eGBTS/NodeB/eNodeB/gNodeB)

It is recommended that the function of automatic switching between different charge-and-discharge modes be enabled if the grid quality is favorable. This helps greatly prolong the battery lifespan.

5.4.1 Data Configuration

5.4.1.1 Data Preparation

Table 5-3 and **Table 5-4** list the parameter settings for intelligent battery management.

Table 5-3 Parameter related to the **EQUIPMENT** MO

Parameter Name	Parameter ID	MAE-Deployment Parameter Name	Setting Notes
Battery Intelligent Management Switch	**EQUIPMENT**.***BATIMS (LTE eNodeB, 5G gNodeB)***	Battery Intelligent Management Switch	This switch can be turned on and take effect only when the license for intelligent battery management has been activated.

Table 5-4 Parameters related to the **BATTERY** MO

Parameter Name	Parameter ID	MAE-Deployment Parameter Name	Setting Notes
Cabinet No.	**BATTERY**.***CN***	Cabinet No.	Set this parameter based on the actual situation.
Subrack No.	**BATTERY**.***SRN***	Subrack No.	Generally, the default value **7** is used.
Slot No.	**BATTERY**.***SN***	Slot No.	Generally, the default value **0** is used.
Lead-acid Battery 1 Capacity	**BATTERY**.***BC1***	Lead-acid Battery 1 Capacity	Set these parameters based on the actual situation. Incorrect settings may affect the lifetime of the batteries. For the requirements on the configuration, see the product manual delivered with the storage batteries.
Temperature Alarm Upper Threshold	**BATTERY**.***TUTHD (LTE eNodeB, 5G gNodeB)***	Temperature Alarm Upper Threshold	This parameter specifies the temperature alarm upper threshold based on the calculation using the following formula: Temperature alarm upper threshold = Float charging trigger threshold under high temperature + 5°C. If the battery operating temperature has been higher than the float charging trigger threshold under high temperature for more than 5 minutes, boost charging is changed to float charging to protect batteries. It is recommended that you use the default value 50°C.

Parameter Name	Parameter ID	MAE-Deployment Parameter Name	Setting Notes
Shutdown Temperature	**BATTERY**.***SDT*** ***(LTE eNodeB, 5G gNodeB)***	Shutdown Temperature	This parameter specifies the high temperature above which the battery is powered off. If the battery operating temperature has been higher than the value of this parameter for more than 5 minutes, the PMU performs one of the following operations to protect batteries: ● Powers off batteries when **High Temperature Shutdown Flag** is set to **ENABLE**. ● Lowers the busbar voltage to high temperature protection voltage (50.5 V) when **High Temperature Shutdown Flag** is set to **DISABLE**. **60** is recommended for the SC48200 PMU. **53** is recommended for other PMUs.
High Temperature Shutdown Flag	**BATTERY**.***HTSDF*** ***(LTE eNodeB, 5G gNodeB)***	High Temperature Shutdown Flag	● This parameter specifies whether to power off batteries under high temperature. When this parameter is set to **ENABLE**, the base station allows battery power-off under high temperature. ● When this parameter is set to **DISABLE**, the base station only lowers the battery charging voltage to ensure power backup under high temperature. ● It is recommended that this parameter be set to **ENABLE** in high-temperature areas.

5.4.1.2 Using MML Commands

Before using MML commands, refer to **5.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

Step 1 Run the **DSP BRDVER** command to query the PMU software version of a base station requiring PMU software activation. If the PMU software is already upgraded to a version that supports this function, PMU software activation is not required.

```
DSP BRDVER: CN=0,SRN=7,SN=0;
```

Step 2 (Optional) Activate the PMU software.

```
//Downloading the version software to an NE
DLD SOFTWARE: MODE=IPV4, IP="192.168.20.48", USR="user", PWD="*****", DIR="/files", SWT=SOFTWARE, SV="BTS3900_5900 V100R016C10SPC100";
//Activating the PMU software
ACT SOFTWARE: OT=BOARD, CN=0, SRN=7, SN=0;
//Checking whether the PMU software has been upgraded to the target version
DSP BRDVER: CN=0,SRN=7,SN=0;
```

Step 3 Set parameters related to intelligent battery management.

```
//Turning on the intelligent battery management switch
SET EQUIPMENT: BATIMS=ON;
//Adding a battery, enabling automatic battery power-off upon high temperature, and setting the power-off temperature and the upper limit of the temperature alarm
ADD BATTERY: CN=0, BC1=600, TUTHD=50, HTSDF=ENABLE, SDT=53, SRN=7, SN=0;
//Enabling automatic battery power-off upon high temperature, and setting the power-off temperature and the upper limit of the temperature alarm
MOD BATTERY: CN=0, TUTHD=50, HTSDF=ENABLE, SDT=53, SRN=7, SN=0;
```

----End

Deactivation Command Examples

Run the **SET EQUIPMENT** command with **EQUIPMENT**.***BATIMS*** set to **OFF**.

```
SET EQUIPMENT: BATIMS=OFF;
```

5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.2 Activation Verification

Intelligent Battery Management

Run the **DSP PMU** command to query PMU status and check whether the value of **Battery Intelligent Management Capability** is **SUPPORTED** and the value of **Battery Intelligent Management Mode** is **ENABLE**. If yes, this function has been activated. If no, this function is not activated.

```
DSP PMU: CN=0;
```

Self-Protection Under High Temperature

Run the **LST BATTERY** command to query battery configurations and check whether the values of **BATTERY**.***HTSDF***, **BATTERY**.***SDT***, and **BATTERY**.***TUTHD*** are the same as the configured ones. If yes, this function has been activated. If no, this function is not activated.

Battery Runtime Display

When the mains supply fails, run the **DSP BATTERY** command and check the value of **Remaining Time** to obtain the remaining power supply duration of storage batteries.

```
DSP BATTERY: CN=0;
```

5.5 Operation and Maintenance (GBTS)

It is recommended that the function of automatic switching between different charge-and-discharge modes be enabled if the grid quality is favorable. This helps greatly prolong the battery lifespan.

5.5.1 Data Configuration

5.5.1.1 Data Preparation

Table 5-5 lists the parameter settings for intelligent battery management.

Table 5-5 Parameter settings for intelligent battery management

Parameter Name	Parameter ID	MAE-Deployment Parameter Name	Setting Notes
Battery Intelligent Management Switch	**BTSOTHPARA.*BATIMS***	Battery Intelligent Management Switch	This switch can be turned on and take effect only when the license for intelligent battery management has been activated.
Temperature Alarm Upper Threshold	**BTSAPMUBP.*TUTHD***	Temperature Alarm Upper Threshold	This parameter specifies the temperature alarm upper threshold based on the calculation using the following formula: Temperature alarm upper threshold = Float charging trigger threshold under high temperature + 5°C. If the battery operating temperature has been higher than the float charging trigger threshold under high temperature for more than 5 minutes, boost charging is changed to float charging to protect batteries. It is recommended that you use the default value 50°C.

Parameter Name	Parameter ID	MAE-Deployment Parameter Name	Setting Notes
Shutdown Temperature	**BTSAPMUBP.** ***SDT***	Shutdown Temperature	This parameter specifies the high temperature above which the battery is powered off. If the battery operating temperature has been higher than the value of this parameter for more than 5 minutes, the PMU performs one of the following operations to protect batteries: ● Powers off batteries when **High Temperature Shutdown Flag** is set to **ENABLE**. ● Lowers the busbar voltage to high temperature protection voltage (50.5 V) when **High Temperature Shutdown Flag** is set to **DISABLE**. ● **60** is recommended for the SC48200 PMU. ● **53** is recommended for other PMUs.
High Temperature Shutdown Flag	**BTSAPMUBP.** ***HTSDF***	High Temperature Shutdown Flag	This parameter specifies whether to power off batteries under high temperature. When this parameter is set to **ENABLE**, the base station allows battery power-off under high temperature. When this parameter is set to **DISABLE**, the base station only lowers the battery charging voltage to ensure power backup under high temperature. It is recommended that this parameter be set to **ENABLE** in high-temperature areas.

5.5.1.2 Using MML Commands

Before using MML commands, refer to 5.2.2 Impacts and 5.3 Requirements and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

Step 1 Perform the following sub-steps to activate the PMU software:

1. Select the base station where the PMU software is to be activated. Run the **DSP BTSBRD** command to query the PMU software version. If the PMU software is already upgraded to a version that supports this function, PMU

software activation is not required. Otherwise, perform subsequent operations.

```
DSP BTSBRD: INFOTYPE=BASEINFO, IDTYPE=BYID, BTSID=10, BRDTYPE=NORMAL, CN=0, SRN=7, SN=0;
```

2. Run the **LOD BTSSW** command to load the PMU software onto the base station.

```
LOD BTSSW: IDTYPE=BYID, BTSIDLST=10, BTSSWVER=BTS3000V100R014C01B027SP17,
BRDSWTYPE=DPMU_MAIN;
```

3. Run the **ACT BTSSW** command to activate the PMU software.

```
ACT BTSSW: IDTYPE=BYID, BTSIDLST=10, VERTYPE=BYBTSSWVER,
BTSSWVER=BTS3000V100R014C01B027SP17, BRDSWTYPE=DPMU_MAIN;
```

4. Run the **DSP BTSBRD** command to verify that the PMU software version meets the minimum version requirements of intelligent battery management.

```
DSP BTSBRD: INFOTYPE=BASEINFO, IDTYPE=BYID, BTSID=10, BRDTYPE=NORMAL, CN=0, SRN=7, SN=0;
```

Step 2 Perform the following sub-steps to set parameters related to this function:

1. Run the **SET BTSOTHPARA** command with **BTSOTHPARA**.***BATIMS*** set to **ON**.

```
SET BTSOTHPARA: IDTYPE=BYID, BTSID=10, BATIMS=ON;
```

2. Run the **SET BTSAPMUBP** command with **BTSAPMUBP**.***HTSDF*** set to **ENABLE**, **BTSAPMUBP**.***SDT*** set to **53**, and **BTSAPMUBP**.***TUTHD*** set to **50**.

```
SET BTSAPMUBP: IDTYPE=BYID, BTSID=10, CN=0, SRN=7, SN=0, CFGFLAG=YES,
CELLTEMPCOMPENABLED=YES, TUTHD=50, HTSDF=ENABLE, SDT=53;
```

----End

Deactivation Command Examples

Run the **SET BTSOTHPARA** command with **BTSOTHPARA**.***BATIMS*** set to **OFF**.

5.5.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.5.2 Activation Verification

Method 1: Run the **DSP BTSBRD** command to query the basic information of each board in the power system and check whether the board status is normal.

Method 2: Check whether power devices such as the PMU, PSU, and PDB are normal on the MAE-Access or Web LMT device panel.

NOTE

The power devices in the PMU subrack, such as the PSU and PDB, are managed by the PMU. The alarm status of these devices is displayed under the PMU.

5.6 Network Monitoring

None

6 Automatic Battery Testing Management

6.1 Principles

An automatic battery test can be performed in a standard or simplified manner. During a test, the following items are checked: battery efficiency, discharge duration, end-of-discharge voltage, discharge average current, and accumulated discharged power (Ah). In this way, battery quality issues can be promptly identified, so that long-time discharging can be avoided and the battery health status can be determined. Automatic battery testing takes effect for lead-acid batteries and the blade lithium battery module IBBS20L.

During a standard test, the battery efficiency will be calculated. The test duration is long because batteries must be discharged until the battery voltage reaches the discharge end voltage. If the mains supply fails during battery discharging, battery runtime cannot reach the predefined level, which affects base station operations.

In a simplified test, the battery efficiency will not be calculated. It is unnecessary to consider the battery discharge time that needs to be obtained by waiting for several hours. In this way, whether the battery is normal can be determined within a short period of time, reducing the test time. A simplified test requires a small depth of discharge (DOD). Therefore, it has a slight impact on the battery lifespan.

The battery characteristic curve differs by battery manufacturer and battery model. In addition, the fluctuation of the load current and the temperature change during discharging affect the test result. The test result is for reference only and is not the standard for replacing the battery.

This function applies to GSM, UMTS, LTE, and NR base stations.

6.1.1 Standard Testing

A standard battery test is performed to determine the battery efficiency, which is used to check whether the batteries are working properly. The formula for calculating the battery efficiency is as follows:

Battery efficiency = Actual discharge duration/Expected discharge duration x 100%

- If battery efficiency < 80%, a battery alarm is reported.

- If 80% ≤ battery efficiency < value of the ***EFF*** parameter, a battery pre-alarm is reported.
- If battery efficiency ≥ value of the ***EFF*** parameter, the batteries are working properly.

Standard automatic battery tests are classified as follows:

- Scheduled automatic test: It is started automatically by the system when the time limit of scheduled discharging test is exceeded and no battery test is performed within the time limit.
- Power-off automatic test: It is started when AC power fails.
- Scheduled and power-off automatic test

NOTE

- The expected discharge duration is related to the average discharge current. The larger the average discharge current, the shorter the expected discharge duration. The PMU dynamically determines the expected discharge duration based on the present discharge current.
- The scheduled automatic test will not be started if any of the following alarms is reported: ALM-25622 Mains Input Out of Range, ALM-25624 Battery Power Unavailable, ALM-25625 Battery Current Out of Range, ALM-25626 Power Module Abnormal, ALM-25630 Power Module and Monitoring Module Communication Failure, ALM-25634 Battery Not In Position, and ALM-25654 Battery Temperature Unacceptable. When the scheduled automatic test has been started, the test will terminate if any of the preceding alarms is reported.
- The power-off automatic test starts when AC power fails and the interval between the present time and the previous test time is greater than the discharge test time limit.
- If the PMU is reset and restarted, the battery test time is restored to zero and the counting starts again.
- Regardless of the test type, PSU insertion during the test will terminate the test.

6.1.2 Simplified Testing

A simplified test is used to determine the battery status based on the end-of-discharge voltage and test time limit. Batteries start discharging after a simplified test starts. The test stops when the battery voltage is lower than the value of the ***SDSEV (LTE eNodeB, 5G gNodeB)*** parameter.

Batteries are properly working if the battery voltage remains higher than the value of the ***SDSEV (LTE eNodeB, 5G gNodeB)*** parameter once the ***SDSTML (LTE eNodeB, 5G gNodeB)*** parameter value is reached during the test.

Generally, the ***SDSTML (LTE eNodeB, 5G gNodeB)*** parameter is set to a value that is equal to the battery backup duration required by the operator or set to a large value. When the parameter value equals the required battery backup duration and the specified discharge time limit expires before the battery voltage reaches the end-of-discharge voltage, the remaining power of the batteries can meet the power backup requirement and the batteries are properly working. When the ***SDSTML (LTE eNodeB, 5G gNodeB)*** parameter is set to a large value, it generally satisfies the requirement of calculating the battery efficiency in a standard test. As a result, the battery voltage reaches the end-of-discharge voltage before the specified discharge time limit expires. Users can then obtain the actual battery discharge duration to calculate battery efficiency. By default, the ***SDSTML (LTE eNodeB, 5G gNodeB)*** parameter is set to a large value.

6.2 Network Analysis

6.2.1 Benefits

None

6.2.2 Impacts

None

6.3 Requirements

6.3.1 Licenses

There are no license requirements for basic functions.

6.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.3.2.1 Prerequisite Functions

None

6.3.2.2 Mutually Exclusive Functions

None

6.3.3 Hardware

Base Station Models

- 3900 and 5900 series base stations (macro base stations)
- DBS3900 LampSite and DBS5900 LampSite

Power Supply Systems

For details, see **Power Supply** and **Mapping Between Physical Cabinets and PTYPE Values**.

6.3.4 Others

This function has no requirements on the base station software version, PMU software version, and power supply systems.

6.4 Operation and Maintenance (eGBTS/NodeB/eNodeB/gNodeB)

This function can be enabled based on requirements for Huawei AC-powered base stations equipped with PMUs.

6.4.1 Data Configuration

6.4.1.1 Data Preparation

Table 6-1 lists the parameter settings for automatic battery testing management.

Table 6-1 Parameters related to the **BATCTPA** MO

Parameter Name	Parameter ID	Setting Notes
0.05C10 Discharge Time 0.1C10 Discharge Time 0.2C10 Discharge Time 0.3C10 Discharge Time 0.4C10 Discharge Time 0.5C10 Discharge Time 0.6C10 Discharge Time 0.7C10 Discharge Time 0.8C10 Discharge Time 0.9C10 Discharge Time	**BATCTPA**.***DSCHGT0 (LTE eNodeB, 5G gNodeB)*** **BATCTPA**.***DSCHGT1 (LTE eNodeB, 5G gNodeB)*** **BATCTPA**.***DSCHGT2 (LTE eNodeB, 5G gNodeB)*** **BATCTPA**.***DSCHGT3 (LTE eNodeB, 5G gNodeB)*** **BATCTPA**.***DSCHGT4 (LTE eNodeB, 5G gNodeB)*** **BATCTPA**.***DSCHGT5 (LTE eNodeB, 5G gNodeB)*** **BATCTPA**.***DSCHGT6 (LTE eNodeB, 5G gNodeB)*** **BATCTPA**.***DSCHGT7 (LTE eNodeB, 5G gNodeB)*** **BATCTPA**.***DSCHGT8 (LTE eNodeB, 5G gNodeB)*** **BATCTPA**.***DSCHGT9 (LTE eNodeB, 5G gNodeB)***	These parameters specify the duration of discharging the battery with a specified discharge current to the moment when the discharging voltage is terminated. If the discharge current is 0.05C10, C10 indicates the nominal capacity of the battery, and 0.05C10 indicates that the battery is discharged with a current of 0.05 x nominal capacity of the battery. For example, if the battery capacity is 100 Ah, 0.05C10 indicates that the battery is discharged with a capacity of 5 Ah per hour. You are advised to set these parameters based on site requirements.
Battery Charge Efficiency	**BATCTPA**.***EFF (LTE eNodeB, 5G gNodeB)***	**80** is recommended.
Discharge Test End Voltage	**BATCTPA**.***ENDV (LTE eNodeB, 5G gNodeB)***	This parameter is one of the criteria for terminating a discharge test. If the battery voltage is lower than the value of this parameter, the battery is completely discharged and the measured results can be recorded. This parameter applies only to a single battery. **190** is recommended.

Parameter Name	Parameter ID	Setting Notes
Battery Number	**BATCTPA**.***BATNUM (LTE eNodeB, 5G gNodeB)***	This parameter specifies the number of storage batteries in a battery group. **24** is recommended.
Discharge Test Time Limit	**BATCTPA**.***DSTML (LTE eNodeB, 5G gNodeB)***	Set this parameter to a value greater than or equal to the battery backup duration required by the operator. **10** is recommended.
Simple Discharge Test End Voltage	**BATCTPA**.***SDSEV (LTE eNodeB, 5G gNodeB)***	In a simplified test, the storage battery's efficiency is not calculated. The **BATCTPA.*SDSEV (LTE eNodeB, 5G gNodeB)*** and **BATCTPA.*SDSTML (LTE eNodeB, 5G gNodeB)*** parameters are used to determine the storage battery's performance. **450** is recommended. When **PMU**.***PTYPE*** is set to **SMU**, **BATCTPA**.***SDSEV*** must be set to a value greater than or equal to **442** and less than or equal to **530**.
Simple Discharge Test Time Limit	**BATCTPA**.***SDSTML (LTE eNodeB, 5G gNodeB)***	This parameter specifies the time limit for a simplified discharge test. **60** is recommended.
Automatic Test Mode	**BATCTPA**.***ATMODE (LTE eNodeB, 5G gNodeB)***	There are four types of test modes: NOAUTOTEST (No Auto Test), PERIOD (Period Auto Test), POWERCUT (Power Off Auto Test), and PERIODPOWERCUT (Period and Power Off Auto Test). **POWERCUT** is recommended.
Timing Discharge Test Time	**BATCTPA**.***TDSTM (LTE eNodeB, 5G gNodeB)***	For a scheduled automatic test, the following parameter settings are involved: ● The **BATCTPA**.***TDSTM*** parameter specifies the interval at which an automatic test is performed, and its value must be greater than that of the **BATCTPA**.***DDSTM*** parameter. The value **120** is recommended. ● The **BATCTPA**.***TDSTM*** parameter specifies the delay for starting an automatic test. The value **14** is recommended.
Delayed Discharge Test Time	**BATCTPA**.***DDSTM (LTE eNodeB, 5G gNodeB)***	

6.4.1.2 Using MML Commands

Before using MML commands, refer to **6.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

Run the **MOD BATCTPA** command to set the parameters associated with automatic battery tests. For details about parameter settings, see **6.4.1.1 Data Preparation**.

Deactivation Command Examples

N/A

6.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.4.2 Activation Verification

To check whether automatic battery testing management has been activated, perform the following steps:

Step 1 Run the **STR BATTST** command to start a simplified or standard test according to customer needs. A standard test is recommended.

- Standard test: Batteries are fully charged before undergoing a discharge test for battery efficiency measurement.
- Simplified test: A discharge test is performed directly.

Step 2 Run the **DSP BATTR** command to query the battery test result. In the displayed test result, check whether the battery efficiency is higher than or equal to 80%. Generally, if the battery efficiency is lower than 80%, the batteries are considered abnormal. In scenarios where lead-acid and lithium batteries are used together for power backup, the battery efficiency displayed in the test result is the efficiency of lead-acid batteries.

Step 3 Run the **STP BATTST** command to stop the test.

----End

6.5 Operation and Maintenance (GBTS)

This function can be enabled based on requirements for Huawei AC-powered base stations equipped with PMUs.

6.5.1 Data Configuration

6.5.1.1 Data Preparation

Table 6-2 lists the parameter settings for automatic battery testing management on the GSM side.

Table 6-2 Parameter settings for automatic battery testing management.

Parameter Name	Parameter ID	Setting Notes
0.05C10 Discharge Time 0.1C10 Discharge Time 0.2C10 Discharge Time 0.3C10 Discharge Time 0.4C10 Discharge Time 0.5C10 Discharge Time 0.6C10 Discharge Time 0.7C10 Discharge Time 0.8C10 Discharge Time 0.9C10 Discharge Time	**BTSAPMUBP**.***DSCHGT0*** **BTSAPMUBP**.***DSCHGT1*** **BTSAPMUBP**.***DSCHGT2*** **BTSAPMUBP**.***DSCHGT3*** **BTSAPMUBP**.***DSCHGT4*** **BTSAPMUBP**.***DSCHGT5*** **BTSAPMUBP**.***DSCHGT6*** **BTSAPMUBP**.***DSCHGT7*** **BTSAPMUBP**.***DSCHGT8*** **BTSAPMUBP**.***DSCHGT9***	These parameters specify the duration of discharging the battery with a specified discharge current to the moment when the discharging voltage is terminated. If the discharge current is 0.05C10, C10 indicates the nominal capacity of the battery, and 0.05C10 indicates that the battery is discharged with a current of 0.05 x nominal capacity of the battery. For example, if the battery capacity is 100 Ah, 0.05C10 indicates that the battery is discharged with a capacity of 5 Ah per hour. Set these parameters based on site requirements.
Battery Charge Efficiency	**BTSAPMUBP**.***EFF***	**80** is recommended.
Discharge Test End Voltage	**BTSAPMUBP**.***ENDV***	**190** is recommended.
Battery Number	**BTSAPMUBP**.***BATNUM***	**24** is recommended.
Discharge Test Time Limit	**BTSAPMUBP**.***DSTML***	Set this parameter to a value greater than or equal to the battery backup duration required by the operator. **10** is recommended.
Simple Discharge Test End Voltage	**BTSAPMUBP**.***SDSEV***	**450** is recommended.

<table>
<tr><th>Parameter Name</th><th>Parameter ID</th><th>Setting Notes</th></tr>
<tr><td>Simple Discharge Test Time Limit</td><td>BTSAPMUBP.SDSTML</td><td>60 is recommended.</td></tr>
<tr><td>Automatic Test Mode</td><td>BTSAPMUBP.ATMODE</td><td>This parameter specifies whether an automatic test is performed and which automatic mode is used.
POWERCUT is recommended.</td></tr>
<tr><td>Timing Discharge Test Time</td><td>BTSAPMUBP.TDSTM</td><td rowspan="2">For a scheduled automatic test, the following parameter settings are involved:
• BTSAPMUBP.TDSTM: which specifies the interval at which an automatic test is performed. Its value must be greater than that of the BTSAPMUBP.DDSTM parameter. The value 120 is recommended.
• BTSAPMUBP.TDSTM: which specifies the delay for starting an automatic test. The value 14 is recommended.</td></tr>
<tr><td>Delayed Discharge Test Time</td><td>BTSAPMUBP.DDSTM</td></tr>
<tr><td>Battery Test Parameter Configure Enabled</td><td>BTSAPMUBP.BTPC</td><td>NO is recommended.</td></tr>
</table>

6.5.1.2 Using MML Commands

Before using MML commands, refer to **6.2.2 Impacts** and **6.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

Run the **SET BTSAPMUBP** command to set the parameters associated with automatic battery tests. For details about parameter settings, see **6.5.1.1 Data Preparation**.

Deactivation Command Examples

N/A

6.5.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.5.2 Activation Verification

To check whether automatic battery testing management has been activated, perform the following steps:

Step 1 Run the **STR BTSBATTTST** command to start a simplified or standard test according to customer needs. A standard test is recommended.

- Standard test: Batteries are fully charged before undergoing a discharge test for battery efficiency measurement.
- Simplified test: A discharge test is performed directly.

Step 2 In the displayed test result, check whether the battery efficiency is higher than or equal to 80%. Generally, if the battery efficiency is lower than 80%, the batteries are considered abnormal.

Step 3 Run the **STP BTSBATTTST** command to stop the test.

----End

6.6 Network Monitoring

None

7 iLiBattScan

7.1 Principles

The state of health (SOH) of lithium-ion batteries gradually degrades over time, and only deep charge/discharge cycles trigger SOH updates. In class 1 and class 2 power-grid environments, lithium-ion batteries do not undergo deep charge/discharge cycles for extended periods. As a result, the lithium-ion battery SOH returned by the system greatly deviates from the actual SOH. To address this issue, iLiBattScan is introduced. This function enables proactive initiation of deep charge/discharge cycles on lithium-ion batteries, improving the timeliness of the SOH returned by the system.

iLiBattScan can be enabled by turning on the function switch and configuring the required parameters. Based on these settings, the power monitoring system performs polling-based voltage adjustment on the configured Huawei voltage-boosting lithium-ion batteries. In polling-based voltage adjustment, the voltage of the battery under test must be greater than both the PSU voltage and the voltage of each battery in the queue. In polling mode, only one battery is tested at a time, while batteries in the queue and those that have been tested remain fully charged. Unlike group-based testing—where all batteries charge and discharge simultaneously—polling minimizes the impact on the backup power duration (although the backup power duration of the battery under test is still affected). Based on the **Intel Lithium Bat Scan Test Time** parameter, intelligent lithium battery capacity verification tests are performed periodically and are started only in the early morning.

iLiBattScan can be configured as follows:

Step 1 Verify function support. Run the **DSP PMU** command to check whether intelligent lithium battery capacity verification tests are supported.

Step 2 Turn on the function switch and configure the required parameters.

- Set **BATCTPA**.***ILIBATTSCANSW (LTE eNodeB, 5G gNodeB)*** to **ON**.
- Set **BATCTPA**.***ILIBATTSCTM (LTE eNodeB, 5G gNodeB)*** to **120**.

----End

NOTICE

- This feature is intended primarily for use in class 1 and class 2 power-grid environments. Enabling this feature is not recommended in class 3 or class 4 power-grid environments. (class 1: the average monthly duration of AC input outages affecting communications equipment is less than 10 hours; class 2: the average weekly duration of AC input outages affecting communications equipment is less than 10 hours; class 3: the average daily duration of AC input outages affecting communications equipment is greater than or equal to 2 hours and less than 8 hours; class 4: the average daily duration of AC input outages affecting communications equipment is greater than 8 hours or mains power is unavailable)
- During the discharge phase of the intelligent lithium battery capacity verification test, the busbar voltage may drop by a maximum of 1 V.
- If only one lithium-ion battery is configured, this feature still works. However, because no other lithium-ion batteries are available for queuing, the polling logic operates the same as group-based testing, and the impact on the backup power duration cannot be reduced.
- After all lithium-ion batteries have completed the test, the updated SOH can be viewed in the test results. The probability of SOH updates decreases when any of the following conditions is met:
  - During the discharge phase of the test, the current is less than 0.1 C.
  - During the discharge phase of the test, the current fluctuation is greater than 0.1 C.
  - During the discharge phase of the test, the electrochemical cell temperature is less than 15°C.
  - The interval since the last SOH update is less than or equal to 30 days.

7.2 Network Analysis

7.2.1 Benefits

With iLiBattScan, the power system supports remote testing of Huawei voltage-boosting lithium-ion batteries.

7.2.2 Impacts

The impacts between this function and other functions are described in **7.3.2.3 Function Impacts**.

This function has no impact on the network.

7.3 Requirements

7.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
MRFD-221264	iLiBattScan (NR)	NR0SILIBAS00	per gNodeB
MRFD-221224	iLiBattScan (LTE FDD)	LT1SILIBAS00	per eNodeB
MRFD-221234	iLiBattScan (LTE TDD)	LT4SILIBAS00	per eNodeB
MRFD-221214	iLiBattScan (UMTS)	LQW9ILBS01	per NodeB
MRFD-221204	iLiBattScan (GSM)	LGB3LDHR	per BTS

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists (FDD)*, *License Control Item Lists (CIoT)*, and *License Control Item Lists (TDD)* in eRAN feature documentation.

In a multimode base station, iLiBattScan takes effect as long as the license for one RAT is available. The license for a higher RAT is recommended.

7.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.3.2.1 Prerequisite Functions

None

7.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference
NR FDD Low-frequency NR TDD High-frequency NR TDD LTE FDD LTE TDD UMTS GSM	PSU intelligent coordinated energy saving	**IPWRM**.***PSUICESS (LTE eNodeB, 5G gNodeB)*** set to **ON**	*Green Site*
NR FDD Low-frequency NR TDD High-frequency NR TDD LTE FDD LTE TDD UMTS GSM	PSU current limiting protection in case of AC circuit breaker capability insufficiency	**PMU**.***ACCLF (LTE eNodeB, 5G gNodeB)*** set to **ENABLE**	*Power Supply Management*

RAT	Function Name	Function Switch	Reference
NR FDD Low-frequency NR TDD High-frequency NR TDD LTE FDD LTE TDD UMTS GSM	Staggering electricity usage	**EQUIPMENT**.***STAGGEREDPOWERSWITCH (5G gNodeB, LTE eNodeB)*** set to **ON**	*Green Site*
NR FDD Low-frequency NR TDD High-frequency NR TDD LTE FDD LTE TDD UMTS GSM	PSU intelligent shutdown	**PSUIS**.***PSUISS (LTE eNodeB, 5G gNodeB)*** set to **ENABLE**	*Green Site*

RAT	Function Name	Function Switch	Reference
NR FDD Low-frequency NR TDD High-frequency NR TDD LTE FDD LTE TDD UMTS GSM	Automatic PSU voltage optimization/ Automatic voltage optimization for intelligent lithium batteries	**PMU**.***BVR (LTE eNodeB, 5G gNodeB)*** set to **ON**	*Green Site*
NR FDD Low-frequency NR TDD High-frequency NR TDD LTE FDD LTE TDD UMTS GSM	SOC level-based low power consumption mode	**IPWRM**.***SOCLVLADJSW (LTE eNodeB, 5G gNodeB)*** set to **ON**	*Green Site*

7.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD High-frequency NR TDD LTE FDD LTE TDD UMTS GSM	Power Backup Perception	None	*Green Site*	iLiBattScan involves charging and discharging the battery. If the power supply is disconnected during the discharge phase, the backup power duration is reduced, causing an alarm related to Power Backup Perception (backup power exhaustion warning) to be reported earlier than expected. This may affect normal base station operation.

7.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations)

Cabinets

- APM cabinets: Only APM5930 (excluding those using the PMU.22 hardware version) and APM5950 series power cabinets are supported.
- MTS cabinets: Only the SMU02C power monitoring module is supported.
- In all power supply scenarios, the configured batteries must be Huawei voltage-boosting lithium-ion batteries, including the ESM-48100C1, ESM-48100B1, ESM-48150B1, ESM-48100A7, ESM-48200B1, ESM-48150A3, ESM-48200A1, and ESM-48100B5.

Boards

- Main control boards: Only the UMPT supports this function.
- Baseband processing units: N/A

RF Modules

N/A

7.3.4 Others

- Before this function is enabled, batteries must have been configured for the base station (by using the **ADD BATTERY** command). **Table 7-1** lists the battery model requirements.
- Before this function is enabled, all the following conditions must be met: (1) The circuit breakers connecting the battery and the cabinet are closed. (2) No battery-related or PMU/PSU-related alarms are reported. (3) All lithium-ion batteries are fully charged.

Table 7-1 Battery model requirements

Battery Type	Battery Model	Support for iLiBattScan
Lithium-ion batteries	ESM-48100C1/ ESM-48100B1/ ESM-48150B1/ ESM-48100A7/ ESM-48200B1/ ESM-48150A3/ ESM-48200A1/ ESM-48100B5	Supported
Lead-acid batteries	N/A	Not supported
Hybrid use of lead-acid and lithium-ion batteries	N/A	Not supported

7.4 Operation and Maintenance

7.4.1 Data Configuration

7.4.1.1 Data Preparation

Table 7-2 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Intel Lithium Bat Scan Sw	**BATCTPA.*ILIBATTSCANSW (LTE eNodeB, 5G gNodeB)***	Set this parameter to **ON** if iLiBattScan is required.

Parameter Name	Parameter ID	Setting Notes
Intel Lithium Bat Scan Test Time	**BATCTPA.*ILIBATTSCTM (LTE eNodeB, 5G gNodeB)***	The value **120** is recommended.

7.4.1.2 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to **7.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

Step 1 (Optional) Activate the PMU software.

```
//Activating the PMU
ACT SOFTWARE: OT=BOARD, CN=0, SRN=7, SN=0;
//Checking whether the PMU software has been upgraded to the target version
DSP BRDVER: CN=0,SRN=7,SN=0;
```

Step 2 Run the **DSP PMU** command to check whether the PMU software supports intelligent lithium battery capacity verification tests. If **Function Support** has a value of **ILIBATTSCANABILITY:Supported** in the command output, the PMU software supports intelligent lithium battery capacity verification tests.

```
DSP PMU: CN=0, SRN=7, SN=0;
```

Step 3 Set **Intel Lithium Bat Scan Sw** to **ON** and correctly configure **Intel Lithium Bat Scan Test Time**.

```
MOD BATCTPA: CN=0, SRN=7, SN=0, ILIBATTSCANSW=ON, ILIBATTSCTM=120;
```

----End

NOTE

- For details about the PMU software upgrade, see *Power Supply Management*.
- In separate-MPT scenarios, this function can be enabled when the two conditions are met on either side: (1) **Intel Lithium Bat Scan Sw** is set to **ON** and (2) **Intel Lithium Bat Scan Test Time** is correctly set. If **Intel Lithium Bat Scan Sw** is set to **ON** on both sides, ensure that **Intel Lithium Bat Scan Test Time** is consistently set on both sides.

Deactivation Command Examples

```
//Setting Intel Lithium Bat Scan Sw to OFF
MOD BATCTPA: CN=0, SRN=7, SN=0, ILIBATTSCANSW=OFF;
```


NOTE

- In separate-MPT scenarios, iLiBattScan can be deactivated for the entire base station only after **Intel Lithium Bat Scan Sw** has been set to **OFF** for all the main control boards where this parameter was previously set to **ON**.
- In APM power supply scenarios, after iLiBattScan is enabled, a test is performed in the early hours of the following morning, and subsequent tests are periodically performed based on the **Intel Lithium Bat Scan Test Time** parameter (with a minimum interval of 30 days). If the scheduled interval has not yet elapsed but you want to run the test again sooner, you can set **Intel Lithium Bat Scan Sw** to **OFF** and then to **ON** to trigger a test in the early hours of the following morning.

7.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.4.2 Activation Verification

The activation verification procedure is as follows:

Step 1 Run the **LST BATCTPA** command and check whether the test parameters are correctly set.

Step 2 Run the **DSP BATTERY** command.

- If **Charging/Discharging State** is **Float-Charging**, the intelligent lithium battery capacity verification test has not started or is in a waiting state.
- If **Charging/Discharging State** indicates equalized charging or discharging, the intelligent lithium battery capacity verification test is running.

Step 3 After the test is complete, run the **DSP BATTR** command to query the test result. The test result is displayed by battery number.

----End

NOTE

- Do not remove or insert the lithium-ion battery communications cable during the intelligent lithium battery capacity verification test. Otherwise, the test may fail and terminate, or the electronic label may not be updated and the test result may become inaccurate.
- If the PMU version is rolled back or related spare parts are replaced during the intelligent lithium battery capacity verification test, lithium-ion batteries may automatically return to normal after remaining in the test state for 12 hours.
- If the scheduled time for the intelligent lithium battery capacity verification test has been reached but no test result is available for the period, the possible causes are as follows:
  - Alarms are reported, indicating abnormalities in the battery, PSU, PMU, mains supply, link, or other related components.
  - The ambient temperature is lower than 15°C before the test.
  - Mutually exclusive features are enabled.
  - The battery is not fully charged.
  - The test is still in progress.

7.4.3 Network Monitoring

This feature ensures basic network performance and does not need to be separately monitored.

8 Reporting of Loss of Power Supply Redundancy

8.1 Principles

Reporting of Loss of Power Supply Redundancy is supported by GSM, UMTS, NR, and LTE base stations.

PSUs in a base station should be configured in N+1 backup mode. If PSUs are not configured in N+1 backup mode, Loss of Power Supply Redundancy will be reported. Operators should add PSUs if this alarm is frequently reported.

The ***PAE (LTE eNodeB, 5G gNodeB)*** parameter specifies whether to enable reporting of this alarm. If the ***PAE (LTE eNodeB, 5G gNodeB)*** parameter is set to **ON**, the base station determines whether to report this alarm based on the PSU output power in non-redundancy mode and the total load power.

Total load power P_{in} is calculated as follows:

- Total load power P_{in} = Total load current x Busbar voltage
- Total load current = Battery charging current + Load current
- Load current = Total load current when the batteries are being discharged

Output power P in non-redundancy mode is calculated as follows:

Output power P in non-redundancy mode = (Total number of PSUs configured – 1) x PSU output power

The PSU output power is provided by the PSU manufacturer.

The principles for reporting and clearing this alarm are as follows:

- P_{in} > P: This alarm is reported.
- $P_{in} \leq$ P: This alarm is cleared.

8.2 Network Analysis

8.2.1 Benefits

None

8.2.2 Impacts

None

8.3 Requirements

8.3.1 Licenses

There are no license requirements for basic functions.

8.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

8.3.2.1 Prerequisite Functions

None

8.3.2.2 Mutually Exclusive Functions

None

8.3.3 Hardware

Base Station Models

- 3900 and 5900 series base stations (macro base stations)
- DBS3900 LampSite and DBS5900 LampSite

Power Supply Systems

For details, see **Power Supply** and **Mapping Between Physical Cabinets and PTYPE Values**.

8.3.4 Others

- Base station software version

  The base station software of GBSS14.0/RAN14.0/eRAN3.0/SRAN8.0/5G RAN1.0 or later is required.
- Power supply system type

4.3.3 Hardware describes whether power supply systems support this function.

- PMU software version

  The software version of the PMU delivered with a Huawei wireless base station must be 128 or later.

8.4 Operation and Maintenance (eGBTS/NodeB/eNodeB/gNodeB)

It is recommended that this function be enabled when a Huawei AC-powered base station equipped with the PMU is used and the customer is concerned about the traffic change. After this function is enabled, the system automatically alerts the customer to the insufficiency of power supply so that the customer can add power modules to adapt to the increase in the traffic.

8.4.1 Data Configuration

8.4.1.1 Data Preparation

Table 8-1 lists the parameter settings for the reporting of Loss of Power Supply Redundancy.

Table 8-1 Parameter settings for the reporting of Loss of Power Supply Redundancy

Parameter Name	Parameter ID	Setting Notes
Power Lose Redundancy Alarm Enabled	**EQUIPMENT**.***PAE (LTE eNodeB, 5G gNodeB)***	The base station will report a Loss of Power Supply Redundancy alarm if this parameter is set to **ON** and PSUs are not configured in N+1 backup mode.

8.4.1.2 Using MML Commands

Before using MML commands, refer to **8.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

Run the **SET EQUIPMENT** command with **EQUIPMENT**.***PAE*** set to **ON**.

Deactivation Command Examples

Run the **SET EQUIPMENT** command with **EQUIPMENT**.***PAE*** set to **OFF**.

8.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

8.4.2 Activation Verification

Remove one PSU when PSUs are configured in N+1 backup mode and, if the base station immediately reports ALM-25636 Loss of Power Supply Redundancy, the function has been activated.

8.5 Operation and Maintenance (GBTS)

It is recommended that this function be enabled when a Huawei AC-powered base station equipped with the PMU is used and the customer is concerned about the traffic change. After this function is enabled, the system automatically alerts the customer to the insufficiency of power supply so that the customer can add power modules to adapt to the increase in the traffic.

8.5.1 Data Configuration

8.5.1.1 Data Preparation

Table 8-2 lists the parameter settings for the reporting of Loss of Power Supply Redundancy.

Table 8-2 Parameter settings for the reporting of the Loss of Power Supply Redundancy Alarm

Parameter Name	Parameter ID	Setting Notes
Power Lose Redundancy Alarm Enabled	**BTSPLRALM**.*PAE*	The base station will report a Loss of Power Supply Redundancy alarm if this parameter is set to **ON** and PSUs are not configured in N+1 backup mode.

8.5.1.2 Using MML Commands

Before using MML commands, refer to **8.2.2 Impacts** and **8.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

Run the **SET BTSPLRALM** command with **BTSPLRALM**.*PAE* set to **YES**.

Deactivation Command Examples

Run the **SET BTSPLRALM** command with **BTSPLRALM**.***PAE*** set to **NO**.

8.5.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

8.5.2 Activation Verification

Remove one PSU when PSUs are configured in N+1 backup mode and, if the base station immediately reports ALM-25636 Loss of Power Supply Redundancy, the function has been activated.

8.6 Network Monitoring

None

9 Diesel Generator Testing Management

9.1 Principles

Base stations supplied with solar power support the diesel generator testing. Users can start a test to check whether a diesel generator is running properly. If the diesel generator is not running properly, the base station reports an alarm indicating a startup failure.

This function applies to GSM, UMTS, NR, and LTE. For GSM, only the eGBTS supports this function.

9.2 Network Analysis

9.2.1 Benefits

None

9.2.2 Impacts

None

9.3 Requirements

9.3.1 Licenses

There are no license requirements for basic functions.

9.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

9.3.2.1 Prerequisite Functions

None

9.3.2.2 Mutually Exclusive Functions

None

9.3.3 Hardware

Base Station Models

- 3900 and 5900 series base stations (macro base stations)
- DBS3900 LampSite and DBS5900 LampSite

Power Supply Systems

For details, see **Power Supply** and **Mapping Between Physical Cabinets and PTYPE Values**.

9.3.4 Others

Diesel generators have been configured.

9.4 Operation and Maintenance

This function can be enabled based on requirements for Huawei base stations supplied with solar power.

9.4.1 Data Configuration

9.4.1.1 Data Preparation

This function does not require configuration of parameters.

9.4.1.2 Using MML Commands

Before using MML commands, refer to **9.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

The diesel generator test can be started only by running MML commands.

Step 1 Run the **STR DIESELGENTST** command to start the diesel generator test on a PMU.

----End

Deactivation Command Examples

Step 1 Run the **STP DIESELGENTST** command to stop the test.

----End

9.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

9.4.2 Activation Verification

If the diesel generator is not running properly, the base station will report ALM-25697 Diesel Generator Startup Failure.

9.5 Network Monitoring

None

10 Intelligent Diesel Generator Management

10.1 Principles

Base stations supplied with solar power support the intelligent diesel generator management. The PMU uses either RS485 (recommended) or dry contact ports to monitor the status, fuel level, and faults of the diesel generator. For details about the monitoring parameters, see section "Configuring Customized Alarms" in *Monitoring Management Feature Parameter Description*.

If the **ICF (*LTE eNodeB, 5G gNodeB)*** parameter is set to **ENABLE**, the PMU automatically controls the diesel generator, reducing the operation cost of the diesel generator.

This function applies to GSM, UMTS, NR, and LTE.

10.2 Network Analysis

10.2.1 Benefits

None

10.2.2 Impacts

None

10.3 Requirements

10.3.1 Licenses

There are no license requirements for basic functions.

10.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

10.3.2.1 Prerequisite Functions

None

10.3.2.2 Mutually Exclusive Functions

None

10.3.3 Hardware

Base Station Models

- 3900 and 5900 series base stations (macro base stations)
- DBS3900 LampSite and DBS5900 LampSite

Power Supply Systems

For details, see **Power Supply** and **Mapping Between Physical Cabinets and PTYPE Values**.

10.3.4 Others

The PMU type must be SC48200.

10.4 Operation and Maintenance (eGBTS/NodeB/eNodeB/gNodeB)

This function can be enabled based on requirements for Huawei base stations supplied with solar power.

10.4.1 Data Configuration

10.4.1.1 Data Preparation

Table 10-1 lists the parameter settings for intelligent diesel generator management.

Table 10-1 Parameter settings for intelligent diesel generator management

Parameter Name	Parameter ID	Setting Notes
Intelligent Control Flag	**DIESELGEN.*ICF*** ***(LTE eNodeB, 5G gNodeB)***	**ENABLE** is recommended.

10.4.1.2 Using MML Commands

Before using MML commands, refer to **10.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

Run the **MOD DIESELGEN** command with the **DIESELGEN.*ICF*** parameter set to **ENABLE**.

Deactivation Command Examples

Run the **MOD DIESELGEN** command with the **DIESELGEN.*ICF*** parameter set to **DISABLE**.

10.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

10.4.2 Activation Verification

Compare the fuel consumed before and after the function activation to know the amount of fuel that has been saved by this function.

10.5 Operation and Maintenance (GBTS)

This function can be enabled based on requirements for Huawei base stations supplied with solar power.

10.5.1 Data Configuration

10.5.1.1 Data Preparation

Table 10-2 lists the parameter settings for intelligent diesel generator management.

Table 10-2 Parameter settings for intelligent diesel generator management

Parameter Name	Parameter ID	Setting Notes
Intelligent Control Flag	**BTSAPMUBP**.*ICF*	**ENABLE** is recommended.

10.5.1.2 Using MML Commands

Before using MML commands, refer to **10.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

Run the **SET BTSAPMUBP** command with **BTSAPMUBP**.*ICF* set to **ENABLE**.

Deactivation Command Examples

Run the **SET BTSAPMUBP** command with **BTSAPMUBP**.*ICF* set to **DISABLE**.

10.5.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

10.6 Activation Verification

Compare the fuel consumed before and after the function activation to know the amount of fuel that has been saved by this function.

10.7 Network Monitoring

None

11 PSU Current Limiting Protection in Case of AC Circuit Breaker Capability Insufficiency

11.1 Principles

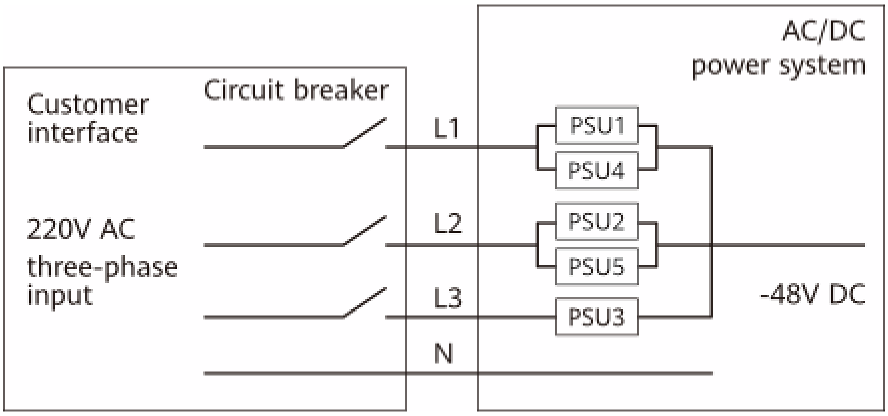
The AC power provided by the customer is converted into DC power by the PSUs and then fed into the main devices of the base station. The customer needs to visit the site to reset the AC circuit breaker after the circuit breaker trips in any of the following scenarios:

- As the power consumption of system loads increases, the PSU output power increases and the AC input current increases accordingly. If there is no increase concerning the AC circuit breaker specification, this customer's circuit breaker will trip.
- In a three-phase AC input scenario, when one PSU in the power system is faulty, the load of the faulty PSU is evenly distributed to normal PSUs. As a result, the output current of the normal PSUs increases, leading to an increase in the input current of phases of normal PSUs, so that the customer's circuit breaker will trip.

ACPT and ***ACASCLT*** are used to restrict the PSU input current below the AC circuit breaker current threshold, thereby preventing the AC circuit breaker from tripping.

For example, in a three-phase AC input scenario, if the specification of the customer's circuit breaker is 32 A and all PSUs are working properly, the current of L1 is 30 A, the current of L2 is 30 A, and the current of L3 is 15 A. If PSU 5 is faulty, the load on the faulty PSU has to be evenly distributed to other PSUs. In this case, the current of L1 is 37.5 A, the current of L2 is 18.75 A, and the current of L3 is 18.75 A. The current of L1 exceeds the upper limit of the circuit breaker provided by the customer, and the customer's circuit breaker trip. If the configured threshold is 32 A, the input current of L1 is limited to 32 A, and the tripping will not occur.

Figure 11-1 Three-phase AC input



This function applies to GSM, UMTS, NR, and LTE. For GSM, only the eGBTS supports this function.

11.2 Network Analysis

11.2.1 Benefits

None

11.2.2 Impacts

- After this function is enabled, if the power required for normal load operation is higher than the maximum output capability of the power system, some loads may be reset due to undervoltage as the load power increases.
- After this function is enabled, the base station may reset repeatedly if the **PMU.*ACASCLT (5G gNodeB, LTE eNodeB)*** parameter value is lower than the power output required for the base station to work properly.
- After this function is enabled, the PSU input current may exceed the **PMU.*ACASCLT (5G gNodeB, LTE eNodeB)*** parameter value if this parameter is set to a small value. This is due to a certain margin of error in the measurement result of the PSU input current. As a result, the restriction on the PSU input current does not take effect.

11.3 Requirements

11.3.1 Licenses

There are no license requirements for basic functions.

11.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

11.3.2.1 Prerequisite Functions

None

11.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference
NR FDD Low-frequency NR TDD High-frequency NR TDD LTE FDD LTE TDD UMTS GSM	iLiBattScan	**BATCTPA**.***ILIBATTSCANSW (LTE eNodeB, 5G gNodeB)*** set to **ON**	*Power Supply Management*

11.3.3 Hardware

Base Station Models

- 3900 and 5900 series base stations (macro base stations)
- DBS3900 LampSite and DBS5900 LampSite

The GSM side must not be a GBTS.

Boards

Only the UMPT supports PSU current limiting protection in case of AC circuit breaker capability insufficiency.

Power Supply Systems

PSU current limiting protection in case of AC circuit breaker capability insufficiency is available only when the power system type is EPS4890, APM, or ETP. For details, see **Power Supply** and **Mapping Between Physical Cabinets and PTYPE Values**.

11.3.4 Others

- Base station software version

  The base station software of RAN22.1/eRAN16.1/SRAN16.1/5G RAN3.1 or later is required.
- Power supply system type

  4.3.3 Hardware describes whether power supply systems support this function.

11.4 Operation and Maintenance

This function can be enabled when the AC circuit breaker configuration does not meet the requirements of the power system.

11.4.1 Data Configuration

11.4.1.1 Data Preparation

Table 11-1 lists the parameter settings for PSU current limiting protection in case of AC circuit breaker capability insufficiency.

Table 11-1 Parameter settings for PSU current limiting protection in case of AC circuit breaker capability insufficiency

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	**PMU**.***CN***	Set this parameter based on the actual situation.
Subrack No.	**PMU**.***SRN***	Generally, the default value **7** is used. For the voltage boosting power distribution subrack, the value range is **17–18**.
Slot No.	**PMU**.***SN***	Generally, the default value **0** is used.

Parameter Name	Parameter ID	Setting Notes
Power System Type	**PMU**.*PTYPE*	● The power supply system type must match the physical cabinet. For the mapping between cabinet types and power supply system types, see **Mapping Between Physical Cabinets and PTYPE Values**. ● When **BATTERY**.*BTYPE* is set to **VRLA_BAT** or **VRLA_BAT_MIX_LI_BAT**, **PMU**.*BVR* must be set to **53.5V** or **DEFAULT** for the PMU that manages the batteries.
AC Current Limiting Flag	**PMU**.*ACCLF (LTE eNodeB, 5G gNodeB)*	**ENABLE** is recommended.
AC Air Switch Current Limiting Threshold	**PMU**.*ACASCLT (LTE eNodeB, 5G gNodeB)*	It is recommended that this parameter be set to the actual capability of the upper-level circuit breaker. The actual capability of the upper-level circuit breaker may be lower than the nominal capability. Therefore, the input current after the AC circuit breaker specifications are derated must be configured based on the actual situation. The circuit breaker derating varies according to the manufacturer and installation position. Therefore, you are advised to check the derating value with the customer before configuration.
AC Phase Type	**PMU**.*ACPT (LTE eNodeB, 5G gNodeB)*	Set this parameter based on the actual situation. It can be set to **SINGLE_PHASE**, **THREE_PHASE**, or **DUAL_LIVE_WIRE**.

11.4.1.2 Using MML Commands

Before using MML commands, refer to **11.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

Step 1 (Optional) Activate the PMU software.

```
//Downloading the version software to an NE
DLD SOFTWARE: MODE=IPV4, IP="192.168.20.48", USR="user", PWD="*****", DIR="/files", SWT=SOFTWARE, SV="BTS3900_5900 V100R016C10SPC100";
```

```
//Activating the PMU software
ACT SOFTWARE: OT=BOARD, CN=0, SRN=7, SN=0;
//Checking whether the PMU software has been upgraded to the target version
DSP BRDVER: CN=0,SRN=7,SN=0;
```

Step 2 Run the **MOD PMU** command. In this step, set parameters as follows:

- **PMU**.***ACCLF*** set to **ENABLE**.
- **PMU**.***ACASCLT*** set to a value based on the input current after the AC circuit breaker specifications of the site are derated, such as **63**
- **PMU**.***ACPT*** set to a value based on the AC input power type of the site, such as **SINGLE_PHASE**.

```
MOD PMU: CN=0, SRN=7, SN=0, PTYPE=ETP, ACCLF=ENABLE, ACASCLT=63, ACPT=SINGLE_PHASE;
```

----End

Deactivation Command Examples

Run the **MOD PMU** command with **PMU**.***ACCLF*** set to **DISABLE**.

11.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

11.4.2 Activation Verification

Run the **DSP PMU** command to query the effective result. The function is successfully activated when the effective result and query result are consistent.

11.5 Network Monitoring

None

12 Hybrid Use of Lead-Acid and Lithium Batteries

12.1 Principles

With the popularization of lithium batteries, lead-acid batteries in the power system are gradually replaced with lithium batteries for power backup. During the replacement, lead-acid and lithium batteries are used together in the power system, and the power system supports both lead-acid and lithium batteries for power backup.

BTYPE (LTE eNodeB, 5G gNodeB) specifies hybrid use of lead-acid and lithium batteries and ***DISCHARGEMODE (LTE eNodeB, 5G gNodeB)*** specifies the battery discharge mode.

This function applies to GSM, UMTS, NR, and LTE. For GSM, only the eGBTS supports this function.

12.2 Network Analysis

12.2.1 Benefits

None

12.2.2 Impacts

After this function is enabled, the power system can use both lead-acid and lithium batteries for power backup.

12.3 Requirements

12.3.1 Licenses

There are no license requirements for basic functions.

12.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

12.3.2.1 Prerequisite Functions

None

12.3.2.2 Mutually Exclusive Functions

None

12.3.3 Hardware

Base Station Models

- 3900 and 5900 series base stations (macro base stations)
- DBS3900 LampSite and DBS5900 LampSite

The GSM side must not be a GBTS.

Boards

Only the UMPT supports the hybrid use of lead-acid and lithium batteries.

Power Supply Systems

- When the power system type is ETP, lead-acid and lithium batteries can be used together. For details, see **Mapping Between Physical Cabinets and PTYPE Values**.
- Lead-acid and lithium batteries have been installed.

12.3.4 Others

- Base station software version

  The base station software of RAN23.0/eRAN17.0/SRAN17.0/5G RAN5.0 or later is required.
- Power supply system type

  4.3.3 Hardware describes whether power supply systems support this function.
- PMU software version

  The requirements for the software version of the PMU delivered with a Huawei radio base station are as follows:

  When the hardware version is PMU.19, the PMU software version must be 219 or later. When the hardware version is PMU.22, the PMU software version must be 213 or later.

12.4 Operation and Maintenance

12.4.1 Data Configuration

12.4.1.1 Data Preparation

Table 12-1 describes the parameters used for the hybrid use of lead-acid and lithium batteries.

Table 12-1 Parameters used for the hybrid use of lead-acid and lithium batteries

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	**BATTERY.*CN (5G gNodeB, LTE eNodeB)***	Set this parameter based on the actual situation.
Subrack No.	**BATTERY.*SRN (5G gNodeB, LTE eNodeB)***	Generally, the default value **7** is used.
Slot No.	**BATTERY.*SN (5G gNodeB, LTE eNodeB)***	Generally, the default value **0** is used.
Battery Type	**BATTERY.*BTYPE (LTE eNodeB, 5G gNodeB)***	• You are advised to set this parameter to **VRLA_BAT_MIX_LI_BAT**. • When ***BTYPE (LTE eNodeB, 5G gNodeB)*** is set to **VRLA_BAT** or **VRLA_BAT_MIX_LI_BAT**, ***BVR*** must be set to **53.5V** or **DEFAULT** for the PMU that manages the batteries.
Discharge Mode	**BATTERY.*DISCHARGEMODE (LTE eNodeB, 5G gNodeB)***	• If this parameter is set to **SIMULTANEOUS_DISCHARGE**, lead-acid and lithium batteries discharge at the same time when the mains supply is faulty. • If this parameter is set to **LI_BAT_PRIMARY**, lithium batteries discharge preferentially when the mains supply is faulty. The discharge time of lead-acid batteries is shortened and their lifespan is prolonged.

12.4.1.2 Using MML Commands

Before using MML commands, refer to **12.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

Step 1 (Optional) Activate the PMU software. (The base station software version must meet the requirements of this function. For details, see **12.3.4 Others**.)

```
//Activating the PMU software
ACT SOFTWARE: OT=BOARD, CN=0, SRN=7, SN=0;
//Checking whether the PMU software has been upgraded to the target version
DSP BRDVER: CN=0,SRN=7,SN=0;
```

Step 2 Run the **MOD BATTERY** command. In this step, set parameters as follows:

- ***BTYPE (LTE eNodeB, 5G gNodeB)*** is set to **VRLA_BAT_MIX_LI_BAT**.
- ***DISCHARGEMODE (LTE eNodeB, 5G gNodeB)*** is set to **SIMULTANEOUS_DISCHARGE**.

```
MOD
BATTERY:CN=0,SRN=7,SN=0,BTYPE=VRLA_BAT_MIX_LI_BAT,DISCHARGEMODE=SIMULTANEOUS_DISCHARGE;
```

----End

Deactivation Command Examples

Run the **MOD BATTERY** command with ***BTYPE (LTE eNodeB, 5G gNodeB)*** set to another value.

12.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

12.4.2 Activation Verification

Run the **DSP BATTERY** command to query the effective result. The function is successfully activated when the effective result and query result are consistent.

12.5 Network Monitoring

None

13 Lithium Battery Software Lock Management

13.1 Principles

In the previous lithium battery anti-theft mechanism, the power system will automatically enable the lithium battery anti-theft function after the power monitoring unit (PMU) sets up the communication with the lithium battery, and the lithium battery anti-theft function cannot be disabled. In some scenarios, the communication between the lithium battery and the PMU may be disconnected. When the disconnection persists beyond the lithium battery software lock timeout threshold, the lithium battery disables the DC output and no longer provides backup power. In addition, a buzzer alarm is generated. After the communication between the lithium battery and the PMU recovers, the alarm is cleared and the lithium battery recovers to provide backup power.

To prevent unexpected reporting of the lithium battery anti-theft alarm in some scenarios (for example, the lithium battery monitoring signal cable is removed or the lithium battery is returned from the site to the warehouse for storage), the power system now supports the function of lithium battery software lock management. A base station–level switch has been added to set the lithium battery software lock flag, which is set to **ENABLE** by default. When the flag is set to **DISABLE**, the lithium battery software lock function is disabled. With this setting, the lithium battery does not check whether its disconnection (if occurred) from the PMU has lasted beyond the lithium battery software lock timeout threshold, and if it has lasted beyond the lithium battery software lock timeout threshold, the lithium battery does not disable the DC output or generate a buzzer alarm.

The **IPWRM**.***LIBATTLOCKFLAG (LTE eNodeB, 5G gNodeB)*** parameter has been added to implement this function. This function applies to GSM, UMTS, NR, and LTE. For GSM, only the eGBTS supports this function.

13.2 Network Analysis

13.2.1 Benefits

None

13.2.2 Impacts

None

13.3 Requirements

13.3.1 Licenses

None

13.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

13.3.2.1 Prerequisite Functions

None

13.3.2.2 Mutually Exclusive Functions

None

13.3.3 Hardware

Base Station Models

3900 and 5900 series base stations (macro base stations)

Cabinets

- APM5950H, APM5950H-L, APM30H (Ver.D), APM30H (Ver.E), APM30H (Ver.F), and APM5930
- BBC5200D-L, BBC5600D, and BBC5600A battery cabinets, and reconstructed IBBS/BBC cabinets housing lithium batteries

Boards

None

Power Supply Systems

- Power supply systems supporting this function are listed in **4.3.3 Hardware**. If the current PMU software version does not meet the requirements described in **Table 4-1**, activate the PMU software of the required version by referring to **13.4.1 Data Configuration**. In addition, the PMU software version must match the SRAN version. For details about the SRAN version requirements of this function, see **13.3.4 Others**.
- The APM30H (Ver.D) and APM30H (Ver.E) cabinets support this function only when the PMU 15A or PMU 15A (Ver.B) is used.
- When a Huawei power supply system is used, this function requires that the lithium battery model be ESM-48100B1, ESM-48150B1, or ESM-48100U2.
- When third-party power supply is used, this function requires that the power monitoring unit be PMU 15A (Ver.B), the power cabinet be either an MTS series one or a power cabinet of another vendor, and the lithium battery model be one of the following: ESM-48100A7 (BOM number: 01076658 or 01076198), ESM-48100B1 (BOM number: 01076631 or 01074746-010), ESM-48100B5 (BOM number: 01076659), ESM-48100C1 (BOM number: 01075414-002), ESM-48100U2 (BOM number: 01075496-005), ESM-48150B1 (BOM number: 01076667 or 01074848-010), ESM-48150A3 (BOM number: 01076670 or 01075632-003), ESM-48200A1 (BOM number: 01076441), and ESM-48200B1 (BOM number: 01076884).
- The MTS series power cabinets require the SMU02C power monitoring unit (BOM number: 02312MML-009).

13.3.4 Others

- The base station software of SRAN19.1 or later is required.
- The PMU 15A (Ver.B) supports the setting of the lithium battery software lock timeout threshold. This setting requires the version be SRAN22.0 or later.
- The SMU supports the setting of the lithium battery software lock timeout threshold. This setting requires the version be SRAN22.1 or later.
- When lithium batteries are used with a PMU at a newly deployed site, the lithium batteries must also be connected to the BBU for management. Otherwise, this function cannot take effect properly.

13.4 Operation and Maintenance

13.4.1 Data Configuration

13.4.1.1 Data Preparation

The following table describes the parameters used for lithium battery software lock management.

Table 13-1 Parameters used for lithium battery software lock management

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	**PMU**.***CN (5G gNodeB, LTE eNodeB)***	Set this parameter based on the actual situation.
Subrack No.	**PMU**.***SRN (5G gNodeB, LTE eNodeB)***	Generally, the default value **7** is used. For the voltage boosting power distribution subrack, the value range is **17–18**.
Slot No.	**PMU**.***SN (5G gNodeB, LTE eNodeB)***	Generally, the default value **0** is used.
Lithium Battery Software Lock Flag	**IPWRM**.***LIBATTLOCKFLAG (LTE eNodeB, 5G gNodeB)***	The value **ENABLE** is recommended.
Lithium Battery Software Lock Timeout	**IPWRM**.***LIBATTLOCKTIMEOUT (LTE eNodeB, 5G gNodeB)***	It is recommended that this parameter be set to **8640** (indicating 144 hours). For the PMU 15A (Ver.B), the value range of this parameter is **10** to **43200**. For other PMUs, the value of this parameter is fixed to **8640** and cannot be configured or modified.

13.4.1.2 Using MML Commands

Before using MML commands, refer to **13.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

Step 1 (Optional) Activate the PMU software. (The base station software version must meet the requirements of this function. For details, see **13.3.4 Others**.)

```
//Activating the PMU software
ACT SOFTWARE: OT=BOARD, CN=0, SRN=7, SN=0;
//Checking whether the PMU software has been upgraded to the target version
DSP BRDVER: CN=0, SRN=7, SN=0;
```

Step 2 Run the **DSP PMU** command to check whether lithium battery software lock management is supported based on the **Li-Battery Software Lock Mgmt Cap** value and whether the lithium battery software lock timeout threshold can be configured based on the **Li Bat Software Lock Timeout Mgmt Capb** value. The execution of step 3 is valid only when lithium battery software lock management is supported.

```
DSP PMU: CN=0, SRN=7, SN=0;
```


Step 3 Run the **SET IPWRM** command with **Lithium Battery Software Lock Flag** set to **ENABLE** and **Lithium Battery Software Lock Timeout** specified (**8640** as an example). The latter parameter can be configured only when the PMU 15A or PMU 15A (Ver.B) is used.

```
SET IPWRM: LIBATTLOCKFLAG=ENABLE, LIBATTLOCKTIMEOUT=8640;
```

NOTE

If the **SET IPWRM** command is executed with **Lithium Battery Software Lock Flag** set to **ENABLE** and **Lithium Battery Software Lock Timeout** left unspecified, the lithium battery software lock timeout threshold is the same as when the latter parameter is set to **4320** for the SMU02C and to **8640** for other PMUs.

----End

Deactivation Command Examples

Run the **SET IPWRM** command with **Lithium Battery Software Lock Flag** set to **DISABLE**.

13.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

13.4.2 Activation Verification

Run the **LST IPWRM** command to query the configuration result. If the value of **Lithium Battery Software Lock Flag** in the MML command output is **Enable**, this function has taken effect.

Run the **LST IPWRM** or **DSP BATTERY** command to check whether the lithium battery software lock timeout threshold is successfully configured.

13.5 Network Monitoring

None

14 Optimized Access Policy on Power Monitoring Modules

14.1 Principles

The base station power system does not support the configuration and management of power monitoring modules like the distributed power unit (DPU), energy control center 500 (ECC500), site control center 800 (SCC800), cabinet power monitor unit 01 (CPMU01), and so on. After such a power monitoring module is added by running the **ADD PMU** command, certain risks may arise. For example, power-module-related parameters may fail to be set, certain power functions may not work, and energy saving level messages may be incorrectly reported. As a result, radio services are affected. To avoid the preceding risks, the access policy on power monitoring modules is optimized.

Assume that **PMU Device Type Check Switch** is set to **ON** and a power monitoring module that is not supported by the power system is connected and configured as a PMU. With this function, the power system identifies that the power monitoring module is not supported, the value of **Availability Status** is **Communication lost** for the corresponding board in the **DSP BRD** command output, ALM-26252 Board Unrecognizable is reported, and users cannot run O&M commands properly.

This function applies to GSM, UMTS, NR, and LTE. For GSM, only the eGBTS supports this function.

14.2 Network Analysis

14.2.1 Benefits

None

14.2.2 Impacts

None

14.3 Requirements

14.3.1 Licenses

None

14.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

14.3.2.1 Prerequisite Functions

None

14.3.2.2 Mutually Exclusive Functions

None

14.3.3 Hardware

Base Station Models

3900 and 5900 series base stations (macro base stations)

Cabinets

None

Boards

None

Power Supply Systems

None

14.3.4 Others

The base station software of SRAN19.1 or later is required.

14.4 Operation and Maintenance

14.4.1 Data Configuration

14.4.1.1 Data Preparation

The following table describes the parameters used for optimizing the access policy on power monitoring modules.

Table 14-1 Parameters used for optimizing the access policy on power monitoring modules

Parameter Name	Parameter ID	Setting Notes
PMU Device Type Check Switch	**IPWRM**.***PMUTYPECHKSW (LTE eNodeB, 5G gNodeB)***	The value **ON(ON)** is recommended.

14.4.1.2 Using MML Commands

Before using MML commands, refer to **14.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

Run the **SET IPWRM** command with **PMU Device Type Check Switch** set to **ON**.

```
SET IPWRM: PMUTYPECHKSW = ON;
```

Deactivation Command Examples

Run the **SET IPWRM** command with **PMU Device Type Check Switch** set to **OFF**.

```
SET IPWRM: PMUTYPECHKSW = OFF;
```

14.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

14.4.2 Activation Verification

Run the **LST IPWRM** command to check whether the configuration has taken effect. If the value of **PMU Device Type Check Switch** is **ON**, this function has taken effect. After a power monitoring module not supported by the power system is connected and configured by running the **ADD PMU** command, you can run the **SCN RS485** command to obtain the scanning result. If the value of **Whether to Allow Access and Management** is **NO**, the function has taken effect.

14.5 Network Monitoring

None

15 Parameters

The following hyperlinked EXCEL files of parameter documents match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- eNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the product documentation delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

Step 1 Open the EXCEL file of parameter reference.

Step 2 On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.

Step 3 Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

Step 1 Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter, which may be only a bit of a parameter. View its information, including the meaning, values, impacts, and product version in which it is activated for use.

----End

16 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- eNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference for the software version used on the live network from the product documentation delivered with that version.

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

Step 1 Open the EXCEL file of performance counter reference.

Step 2 On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.

Step 3 Click **OK**. All counters related to the feature are displayed.

----End

17 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

18 Reference Documents

1. ICC100-N5 Solar Controller User Manual
2. BBU3910C Installation Guide